

DATE OF ISSUE	27.05.11	31.05.11	23.11.11	04.12.13	31.10.14	17.06.15	18.05.16	04.05.17	02.04.25
DRAWING PACKAGE VERSION	3	4	5	6	7	8	9	10	11

GENERAL

M8398-G1	SITE SPECIFICATIONS	B	B	AB	-	A	AB	AB	AB	A
M8398-G2	OVERALL SITE PLAN	B	B	AB	-	A	AB	AB	AB	A
M8398-G3	SITE LAYOUT AND SETOUT PLAN	B	B	AB	-	A	AB	AB	AB	A
M8398-G3-2	ANTENNA AND RRU SETOUT PLAN	-	-	-	-	A	AB	AB	AB	A
M8398-G3-3	RRU SETOUT PLAN	-	-	-	-	-	-	-	-	A
M8398-G4	SITE ELEVATION	-	-	AB	-	A	AB	AB	AB	A

ANTENNAS & TRANSMISSION

M8398-A1	eJV ANTENNA SYSTEM CONFIGURATION - SHEET 1 OF 2	A	A	AB	-	A	AB	AB	AB1	A
M8398-A2	eJV ANTENNA SYSTEM CONFIGURATION - SHEET 2 OF 2	-	-	-	-	-	-	-	-	A
M8398-A3	PHYSICAL ASSET SUMMARY TABLE	-	-	-	-	-	-	-	-	A
M8398-P1	OPTUS RF PLUMBING DIAGRAM	-	-	-	-	-	-	A	AB	A
M8398-P2	OPTUS & VODAFONE RF PLUMBING DIAGRAM	-	-	-	-	-	-	-	-	A
M8398-T1	SITE TRANSMISSION DETAILS	A	A	AB	-	-	-	AB1	AB1	-

STRUCTURAL

M8398-S1	STRUCTURAL STEELWORK NOTES	A	A	AB	-	-	-	-	-	-
M8398-S2	SHELTER SUPPORT PLATFORM DETAILS SHEET 1	A	A	AB	-	-	-	-	-	-
M8398-S3	SHELTER SUPPORT PLATFORM DETAILS SHEET 2	A	A	AB	-	-	-	-	-	-
M8398-S4	ANTENNAS MOUNT DETAILS SHEET 1	A	A	AB	-	A	AB	AB	AB	A
M8398-S5	ANTENNAS MOUNT DETAILS SHEET 2	A	A	AB	-	-	-	-	-	-
M8398-S6	ANTENNAS MOUNT DETAILS SHEET 3	A	A	AB	-	-	-	-	-	-
M8398-S7	ANTENNAS MOUNT DETAILS SHEET 4	A	A	AB	-	-	-	-	-	-
M8398-S8	RRU MOUNT DETAILS	-	-	-	-	A	AB	AB	AB	A
M8398-S9	ANTENNA TURRET MOUNT DETAIL SHEET 1 OF 3	-	-	-	-	A	AB	AB	AB	A
M8398-S10	ANTENNA TURRET MOUNT DETAIL SHEET 2 OF 3	-	-	-	-	A	AB	AB	AB	-
M8398-S11	ANTENNA TURRET MOUNT DETAIL SHEET 3 OF 3	-	-	-	-	A	AB	AB	AB	A

ELECTRICAL

M8398-E1	ELECTRICAL SPECIFICATIONS	-	-	-	-	A	AB	AB	AB	A
M8398-E2	SINGLE LINE DIAGRAM	-	-	-	-	-	-	-	-	A

SHELTER / FITOUT

M8398-F1	EQUIPMENT SHELTER LAYOUT PLAN	A	A	AB	AB	A	AB	AB	AB	A
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EME EXCLUSION ZONES

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LEASE / LICENCE

M8398-L1	MICROWAVE LICENCE APPLICATION	A	A	AB	-	-	-	-	-	-
30084010A	SITE PLAN FOR LEASE PURPOSES	-	-	-	-	A	-	-	-	-

DISTRIBUTION

OPTUS	SUHAIB OBAID	-	-	-	-	-	-	-	-	1
SERVICE STREAM	SAMINA TABASSUM	-	-	-	-	-	-	-	-	1



OPTUS SITE - M8398

DOREEN TOWNSHIP-O

JOINT VENTURE SITE ID: JM6015

INDARA SITE ID: 3501475

DOREEN TOWNSHIP

95 HAZEL GLEN DRIVE

DOREEN, VIC 3754

5G UPGRADE (eJV-OH)

OPTUS WORK AUTHORITY N° 540268



ServiceStream

FOR CONSTRUCTION

Drawing No.
M8398 - 00

DATE OF ISSUE	27.05.11	31.05.11	23.11.11	04.12.13	31.10.14	17.06.15	18.05.16	04.05.17	02.04.25
DRAWING PACKAGE VERSION	3	4	5	6	7	8	9	10	11

REFERENCE DOCUMENTS

OSD-100	STANDARD CONSTRUCTION NOTES	-	-	-	-	-	-	-	-	C
OSD-171-2	SITE SIGNAGE TYPICAL ROOFTOP SITE - PLAN	-	-	-	-	-	-	-	-	B
OSD-171-3	SITE SIGNAGE LEGEND AND NOTES	-	-	-	-	-	-	-	-	B
OSD-900	EQUIPMENT SHELTER FLOOR PLAN (STANDARD LAYOUT)	-	-	-	-	A	-	-	-	-
VPL-SC-M8398	VPL STRUCTURAL CERTIFICATION	-	-	-	-	A	-	-	-	-
CERT_M8398	SUPPLIER STRUCTURAL DESIGN CERTIFICATE - ROOFTOP									
	FOR INDARA INFRASTRUCTURE & ENGINEERING DATED 02-04-2025	-	-	-	-	-	-	-	-	0



OPTUS SITE - M8398

DOREEN TOWNSHIP-O

JOINT VENTURE SITE ID: JM6015

INDARA SITE ID: 3501475

DOREEN TOWNSHIP

95 HAZEL GLEN DRIVE

DOREEN, VIC 3754

5G UPGRADE (eJV-OH)

OPTUS WORK AUTHORITY N° 540268



DISTRIBUTION

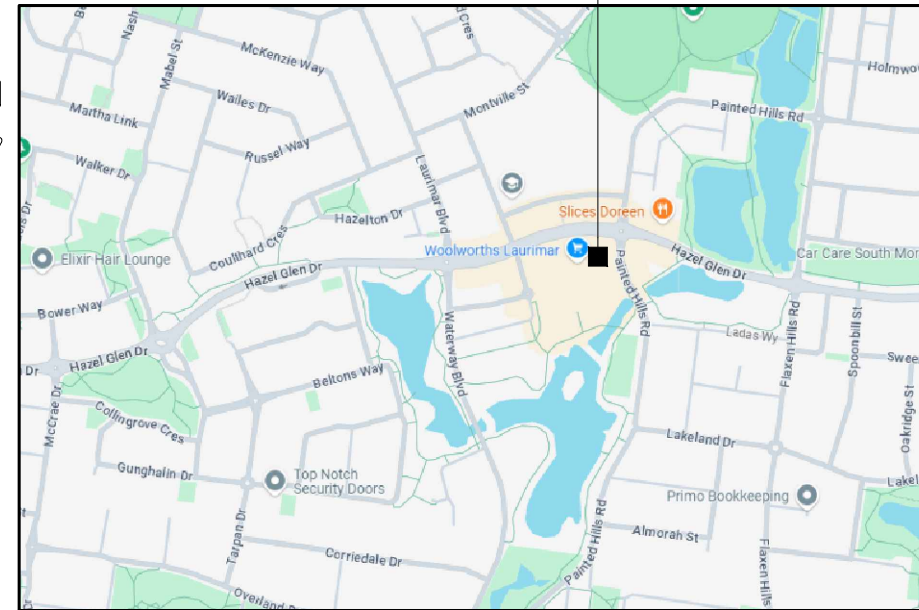
OPTUS	SUHAIB OBAID	-	-	-	-	-	-	-	-	1
SERVICE STREAM	SAMINA TABASSUM	-	-	-	-	-	-	-	-	1

FOR CONSTRUCTION

Drawing No.
M8398 - 01

DOREEN TOWNSHIP
95 HAZEL GLEN DRIVE
DOREEN, VIC 3754

OPTUS SITE M8398
(RFNSA NO: 3754005)



TOPOGRAPHIC MAP ... COPYRIGHT © GOOGLE MAPS

SOURCE: RFNSA

DATUM: MGA (GDA94)	ZONE: 55
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REF LOCATION:	6 ROOFTOP
---------------	-----------

EASTING	334 713
---------	---------

NORTHING	5 838 372
----------	-----------

LATITUDE	-37.58688
----------	-----------

LONGITUDE	145.12794
-----------	-----------

WGS84 DATUM (USED BY GPS DEVICES) CAN BE
CONSIDERED SAME AS GDA94 (SOURCE:
"GEOCENTRIC DATUM OF AUSTRALIA TECHNICAL
MANUAL" VERSION 2.3)

1. EXISTING ANTENNA MOUNTING POLE WITH TURRET MOUNT ON 8.65m HIGH BUILDING PARAPET WALL.
2. NEW PANEL ANTENNAS TO BE INSTALLED ON EXISTING TURRET ON EXISTING MOUNTING POLE. NEW RRU'S TO BE INSTALLED ON EXISTING RRU MOUNT CONNECTED TO CONCRETE WALL.
3. STRUCTURAL ADEQUACY OF EXISTING ANTENNAS & RRU'S MOUNTS HAS BEEN CONFIRMED AND CERTIFIED BY ALLTELECOS. REFER TO CERT_M8398 DATED ON 02/04/2025.
4. ANTENNA MAINTENANCE ACCESS BY QUALIFIED PERSONNEL ONLY VIA LADDER AND STEP PEGS WITH FALL ARREST SYSTEM PROVIDED ON ROOFTOP & MOUNTING POLE.

EXISTING OPTUS PHASE 8 (3m x 2.5m) SANDWICH PANEL EQUIPMENT SHELTER, COLOURED 'WHITE' SUPPORTED ON STEEL FRAME. REFER TO M8398-F1 FOR FURTHER DETAILS.

THIS SITE IS LINKED TO THE NETWORK VIA EXISTING FIBRE.

1. ACCESS VIA HAZEL GLEN DRIVE.
2. ZAP FITNESS TRAVELLING NORTH ON YAN YEAN ROAD TURN LEFT INTO HAZEL GLEN ROAD AND TRAVEL FOR 1.2km TO 3RD ROUND ABOUT WHERE SITE IS ON LEFT HAND SIDE.
3. YOU NEED TO ACCESS 2 KEYS SAFES. 1ST KEY SAFE - IS TO THE RIGHT OF THE WOOLWORTHS ACCESS BOTTOM RIGHT - FOR KEY SAFE CODE CALL SMC INDARA - IT HAS THE KEY FOR THE ROOF DOOR. 2ND KEY SAFE - IS THE LEFT OF THE ZAP FITNESS ENTRY DOOR FOR CODE CALL SMC INDARA - HAS SWIPE CARD TO GROUND FLOOR GYM DOOR.
4. ALL CONTRACTORS ARE KINDLY REQUESTED TO BOOK THEIR VISIT WITH JUN AT LEAST 24 HOURS IN ADVANCE BEFORE GOING TO THE SITE. YOU CAN REACH JUN AT 0401 347 077 OR VIA EMAIL AT jun.na@fitlg.com. LADSAFE TAGGED OUT OF SERVICE AS LAD SAFE CABLE ISN'T THROUGH TOP BRACKET ENOUGH. TAKE NOTE OF HAZARD AS YOU EXIT ACCESS DOOR AS EDGE OF ROOF IS WITHIN 2 METERS - NO HANDRAIL FOR ACCESS 72 HOURS PRIOR NOTIFICATION MUST BE PROVIDED TO INDARA SITE MANAGEMENT CENTRE BY EMAIL accesstosite@accesstosite.com.au OR FOR EMERGENCY ACCESS BY PHONE 1800 006 667.

1. EXISTING EME TRANSMITTING ANTENNAS
2. MANUAL HANDLING
3. WORKING AT HEIGHTS
4. SLIP, TRIP AND FALLS
5. ELECTRICAL HAZARDS
6. WEATHER / LIGHTNING
7. SUN EXPOSURE
8. UNPROTECTED ROOF EDGES
9. TRAFFIC MANAGEMENT

1. SITE SIGNAGE SHALL BE IN ACCORDANCE WITH OSD-171-2 (ROOFTOP SITE) AND OSD-171-3 (EME SIGNAGE).
2. REMOVE EXISTING MERCS#1 SIGNAGE FROM SHELTER.
3. REMOVE EXISTING MERCS#2 SIGNAGE FROM SHELTER ACCESS DOOR.
4. REMOVE EXISTING MERCS#3 SIGNAGE FROM ROOFTOP ACCESS DOOR.
5. MERCS#2 SIGNAGE TO BE INSTALLED ON ROOFTOP ACCESS DOOR WITH RFNSA NUMBER HARD-STAMPED.
6. EXISTING OUTDATED SITE ENQUIRY SIGN ON SHELTER ACCESS DOOR TO BE REPLACED WITH NEW SIGNAGE.
7. EXISTING FADED HAZARDOUS VOLTAGE SIGN ON METER BOX TO BE REPLACED WITH NEW SIGNAGE.
8. MERCS#5 SIGNAGE TO BE INSTALLED ON BACK OF NEW PANEL ANTENNAS.

REFER TO RADIO COMMUNICATIONS SITE MANAGEMENT BOOK (RCSMB) FOR LATEST EME EXCLUSION ZONES FOR NEW ANTENNAS AT THIS SITE.

EXISTING 40A 3-PHASE AC POWER SUPPLY IS NOT SUFFICIENT FOR THE PROPOSED 5G UPGRADE PROJECT AND REQUIRES TO BE UPGRADED TO 63A 3-PHASE. REFER TO DRGS. M8398-E1 AND E2 FOR ELECTRICAL SPECIFICATIONS AND SINGLE LINE DIAGRAM.

[illegible]

Client:	
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OPTUS

Project:

Project: MOBILE NETWORK
AUSTRALIA
SITE No:- M8398
DOREEN TOWNSHIP-O
95 HAZEL GLEN DRIVE

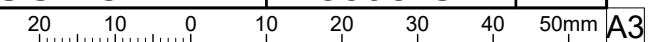
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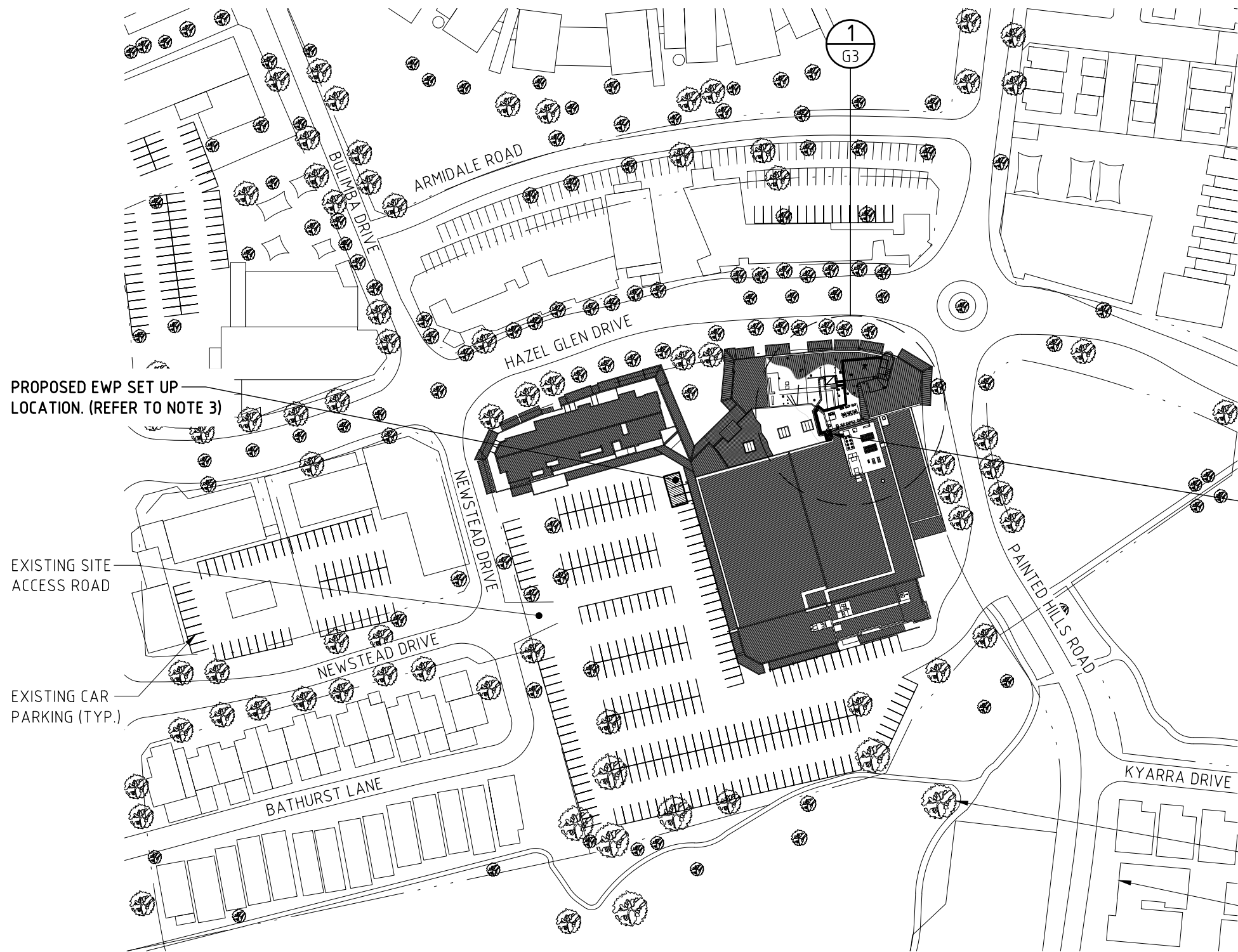
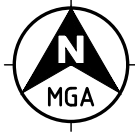
SITE SPECIFICATIONS

Drawing Status:
FOR CONSTRUCTION

Drawing No.
M8398-G1

Revision





NOTES:

- 1. ALL INFORMATION TO BE CHECKED ON SITE PRIOR TO FABRICATION AND CONSTRUCTION.
- 2. DRAWINGS BASED ON INFORMATION PROVIDED BY OTHERS.
- 3. CONSTRUCTION CONTRACTOR TO CONFIRM SUITABILITY OF PROPOSED EWP SET-UP/PARKING LOCATION ON SITE PRIOR TO WORK COMMENCING.
- 4. SERVICES INFORMATION CONTAINED ON THIS DRAWING IS INDICATIVE ONLY AND REFERENCE SHOULD BE MADE TO THE AUTHORITIES DRAWINGS TO CONFIRM ACCURACY AND COMPLETENESS. WHERE INFORMATION IS AVAILABLE, THE SUB-SURFACE SERVICES INSTALLED BY AGENTS OTHER THAN AUTHORITIES HAVE BEEN SHOWN, BUT ADDITIONAL UNDOCUMENTED SERVICES MAY BE PRESENT. SHOULD THE CONTRACTOR BELIEVE THAT SUB-SURFACE SERVICES ARE AT RISK OF DAMAGE DURING CONSTRUCTION, THE CONTRACTOR SHOULD NOTIFY THE RELEVANT AUTHORITIES AND ESTABLISH THE EXACT LOCATION OF THE SERVICES.

LEGEND

- EXISTING PROPERTY BOUNDARY
- EXISTING LEASE AREA
- EXISTING OPTUS U/G POWER SUPPLY

PROPOSED EWP SET UP LOCATION. (REFER TO NOTE 3)

EXISTING SITE ACCESS ROAD

EXISTING CAR PARKING (TYP.)

EXISTING/NEW OPTUS/VODAFONE INSTALLATION. REFER TO DRG. M8398-G3 FOR DETAILS

EXISTING TREES (TYP.)

EXISTING BUILDING (TYP.)

OVERALL SITE PLAN

SCALE 1:1500

Rev	Date	Revision Details	Consultant	CAD	Designer	Verifier	Approver
A	02.04.25	ISSUED FOR CONSTRUCTION (5G UPGRADE (eJV-OHI))	ALL TELECOS	TE	BC	AQ	HR
AB	17.06.15	AS BUILT ISSUED	VPL	RB	MA	RB	NR
A	31.10.14	ISSUED FOR CONSTRUCTION	VPL	PLB	MA	SM	DP
AB	23.11.11	AS BUILT ISSUED	AURECON	SB	NP	SZ	DP
B	09.12.10	ISSUED FOR CONSTRUCTION	AURECON	JND	NP	SZ	DP
A	22.10.10	ISSUED FOR CONSTRUCTION	AURECON	PC	JL		



Client:

OPTUS

Project:

MOBILE NETWORK AUSTRALIA
SITE No:- M8398
DOREEN TOWNSHIP-O
95 HAZEL GLEN DRIVE

Drawing Title:

OVERALL SITE PLAN

Drawing Status:

FOR CONSTRUCTION

Drawing No.

M8398-G2

Revision

A



EXISTING OPTUS PHASE 8 EQUIPMENT SHELTER TO BE RE-CONFIGURED TO ACCOMMODATE EXISTING AND NEW OPTUS/TPG EQUIPMENT. REFER TO DRG. M8398-F1 FOR DETAILS

INSTALL NEW VODAFONE NOKIA FYGC GPS ANTENNA (1 OFF) ON TOP OF SHELTER. MAINTAIN 500mm SEPARATION FROM OPTUS GPS ANTENNA

REPLACE EXISTING OPTUS GPS ANTENNA (1 OFF) WITH NEW OPTUS ERICSSON GNSS GPS ANTENNA (1 OFF) ON TOP OF SHELTER

REMOVE EXISTING MERCS#1 SIGNAGE FROM SHELTER

REMOVE EXISTING MERCS#2 SIGNAGE FROM SHELTER ACCESS DOOR

EXISTING 50mm GAS PIPE OVER ROOF

EXISTING OUTDATED SITE ENQUIRY SIGN ON SHELTER ACCESS DOOR TO BE REPLACED WITH NEW SIGNAGE

EXISTING UPPER ROOF ACCESS LADDER WITH LADSAFE CABLE

EXISTING MERCS#1 SIGNAGE BEHIND ACCESS LADDER

EXISTING OPTUS HANDRAIL
EXISTING OPTUS 600mm WIDE WALKWAY

EXISTING OPTUS 450mm WIDE CABLE TRAY FIXED TO ROOF

EXISTING OPTUS 600mm WIDE STEPOVER FOR CABLE TRAY

EXISTING OPTUS 600mm WIDE WALKWAY

EXISTING OPTUS HANDRAIL AROUND 3 SIDES OF WALKWAY

MGA ZONE 55
E 334 713
N 5 838 372
AT 6 ROOFTOP

EXISTING ACCESS DOOR ONTO LOWER ROOF LEVEL

EXISTING ROOF ACCESS WALKWAY

REMOVE EXISTING MERCS#3 SIGNAGE FROM ROOFTOP ACCESS DOOR

NEW MERCS#2 SIGNAGE TO BE INSTALLED ON ROOFTOP ACCESS DOOR WITH RFNSA NUMBER HARD-STAMPED

EXISTING FADED HAZARDOUS VOLTAGE SIGN ON METER BOX TO BE REPLACED WITH NEW SIGNAGE

EXISTING LIFT & STAIR WELL

EXISTING OPTUS POWER ROUTE

EXISTING BUILDING SUB-MSB ON GROUND FLOOR BEHIND STAIR WELL

EXISTING AIR CONDITIONING UNITS (TYP.)

NOTES:

1. ALL ANTENNA ORIENTATIONS ARE IN DEGREE RELATIVE TO TRUE NORTH.
2. REFER TO DRG. M8398-A1 TO A3 FOR ANTENNA SYSTEM CONFIGURATIONS.
3. REFER TO DRG. M8398-G1 FOR STRUCTURAL ASSESSMENT NOTES.
4. EXISTING CORRODED, DAMAGED OR MISSING ANTENNA MOUNT U-BOLTS AND FIXING TO BE REPLACED AS REQUIRED.
5. IT IS NOTED THAT THE EXISTING SITE E3 DRAWING IS MISSING. IN THE ABSENCE OF AN E3 DRAWING, THE DESIGNER STATES THAT THE PROPOSED OPTUS UPGRADE WORKS WILL BE IN ACCORDANCE WITH RELEVANT SECTIONS OF THE OSD-020 OPTUS EARTHING SPECIFICATION, IN RELATION TO:
 - i) THE NEW EQUIPMENT AND ANY STEELWORK INSTALLED AS PART OF THE SITE UPGRADE WORKS,
 - ii) ANY ALTERATIONS TO THE EXISTING EARTHING SYSTEM.
 - iii) AND THE DESIGNER CONFIRMS ANY SUCH EARTHING WORKS WILL BE FIT FOR PURPOSE.

LEGEND

- EXISTING PROPERTY BOUNDARY
----- EXISTING LEASE AREA
0 e 0 e EXISTING OPTUS U/G POWER SUPPLY

ANTENNA LEGEND



DETAIL SCALE 1:200

2
G3-2

1
G2

Rev	Date	Revision Details	Consultant	CAD	Designer	Verifier	Approver
A	02.04.25	ISSUED FOR CONSTRUCTION (5G UPGRADE (eJV-OH))	ALL TELECOMS	TE	BC	AQ	HR
AB	17.06.15	AS BUILT ISSUED	VPL	RB	MA	RB	NR
A	31.10.14	ISSUED FOR CONSTRUCTION	VPL	PLB	MA	SM	DP
AB	23.11.11	AS BUILT ISSUED	AURECON	SB	NP	SZ	DP
B	09.12.10	ISSUED FOR CONSTRUCTION	AURECON	JND	NP	SZ	DP
A	22.10.10	ISSUED FOR CONSTRUCTION	AURECON	PC	JL		

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Client:

OPTUS

Project:

MOBILE NETWORK AUSTRALIA
SITE No:- M8398
DOREEN TOWNSHIP-O
95 HAZEL GLEN DRIVE

Drawing Title:

SITE LAYOUT AND SETOUT PLAN

Drawing Status:

FOR CONSTRUCTION

Drawing No.

M8398-G3

Revision

A

20 10 0 10 20 30 40 50mm A3



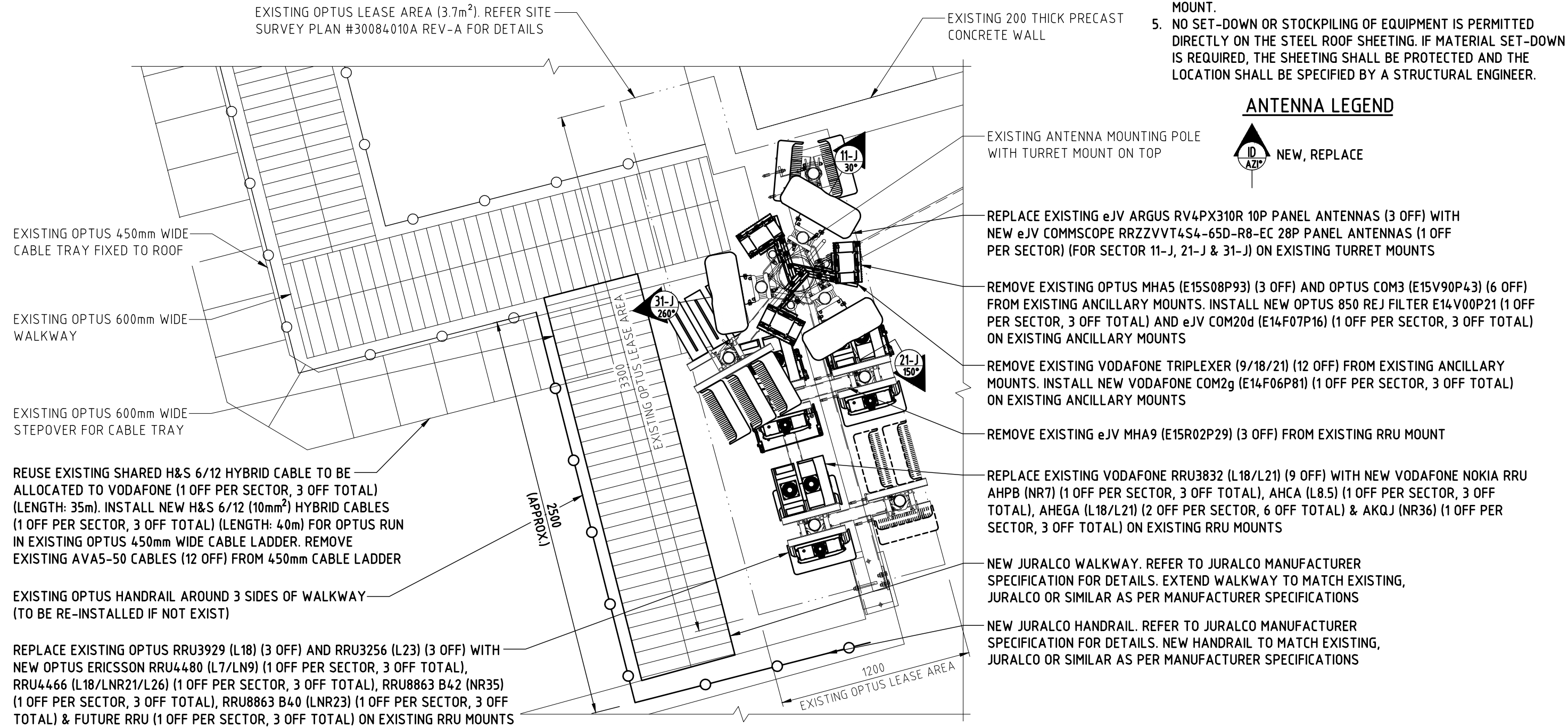
LEGEND

EXISTING LEASE AREA

NOTES:

1. ALL ANTENNA ORIENTATIONS ARE IN DEGREE RELATIVE TO TRUE NORTH.
2. REFER TO DRG. M8398-A1 TO A3 FOR ANTENNA SYSTEM CONFIGURATIONS.
3. REFER TO DRG. M8398-G1 FOR STRUCTURAL ASSESSMENT NOTES.
4. ALL ANCILLARIES TO BE INSTALLED ON EXISTING ANCILLARY MOUNT.
5. NO SET-DOWN OR STOCKPILING OF EQUIPMENT IS PERMITTED DIRECTLY ON THE STEEL ROOF SHEETING. IF MATERIAL SET-DOWN IS REQUIRED, THE SHEETING SHALL BE PROTECTED AND THE LOCATION SHALL BE SPECIFIED BY A STRUCTURAL ENGINEER.

ANTENNA LEGEND



DETAIL SCALE 1:25 2 G3

ANTENNA AND RRU LAYOUT PLAN

Rev	Date	Revision Details	Consultant	CAD	Designer	Verifier	Approver
A	02.04.25	ISSUED FOR CONSTRUCTION (5G UPGRADE (eJV-OHI))	ALL TELECOMS	TE	BC	AQ	HR
AB	17.06.15	AS BUILT ISSUED	VPL	RB	MA	RB	NR
A	31.10.14	ISSUED FOR CONSTRUCTION	VPL	PLB	MA	SM	DP



OPTUS

Client:
Project: MOBILE NETWORK AUSTRALIA
SITE No:- M8398
DOREEN TOWNSHIP-O
95 HAZEL GLEN DRIVE

Drawing Title:
ANTENNA AND RRU SETOUT PLAN
Drawing Status:
FOR CONSTRUCTION
Drawing No.
M8398-G3-2
Revision
A

NOTES:

- 1. REFER TO DRG. M8398-A1 TO A3 FOR ANTENNA SYSTEM CONFIGURATIONS.
- 2. REFER TO DRG. M8398-G1 FOR STRUCTURAL ASSESSMENT NOTES.
- 3. ALL ANCILLARIES TO BE INSTALLED ON EXISTING ANCILLARY MOUNT.
- 4. EXISTING CORRODED, DAMAGED OR MISSING ANTENNA MOUNT U-BOLTS AND FIXING TO BE REPLACED AS REQUIRED.

EXISTING ANTENNA MOUNTING POLE
WITH TURRET MOUNT ON TOP

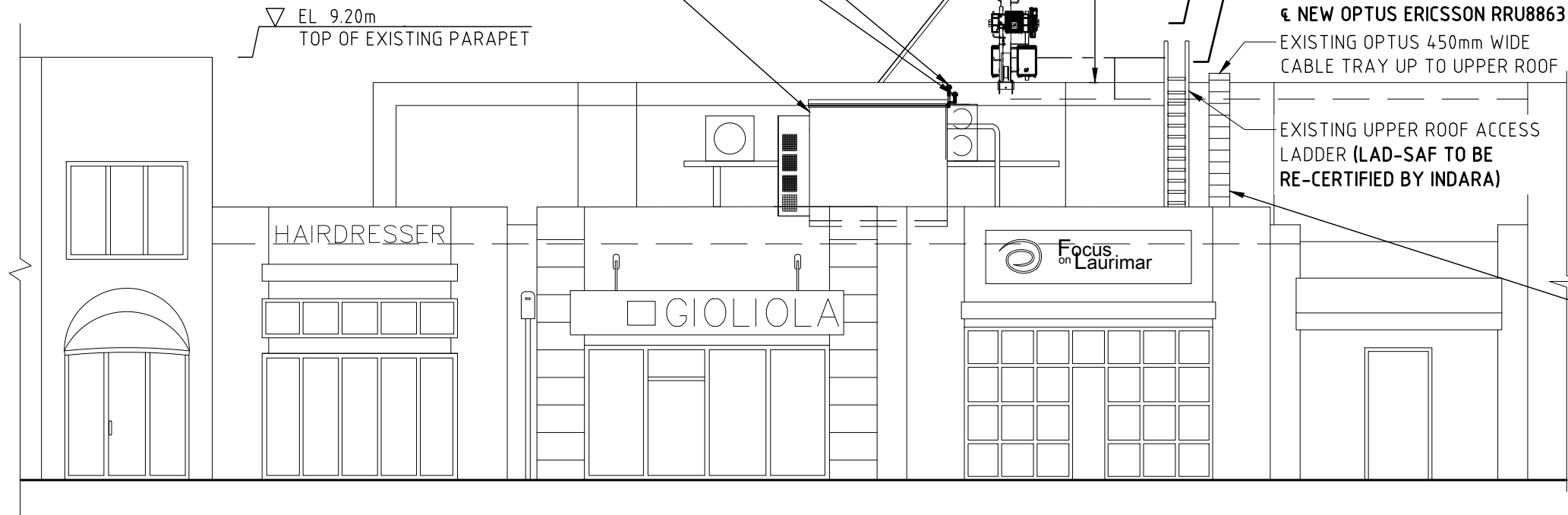
EXISTING OPTUS BACK BRACE

AFTER RECOVERING THE EXISTING FEEDERS RUNNING INSIDE THE
MAIN POLE, BUILD TEAM TO INSTALL MAXIMUM POSSIBLE NO OF
NEW TAILS TO RUN INSIDE THE MAIN POLE TO REDUCE WIND LOAD
& TO ENHANSE ASTHATIC APPERANCE

REPLACE EXISTING OPTUS GPS ANTENNA (1 OFF) WITH NEW OPTUS
ERICSSON GNSS GPS ANTENNA (1 OFF) ON TOP OF SHELTER

INSTALL NEW VODAFONE NOKIA FYGC GPS ANTENNA (1 OFF) ON TOP OF
SHELTER. MAINTAIN 500mm SEPARATION FROM OPTUS GPS ANTENNA

EXISTING OPTUS PHASE 8 EQUIPMENT SHELTER TO BE RE-CONFIGURED
TO ACCOMMODATE EXISTING AND NEW OPTUS/TPG EQUIPMENT. REFER
TO DRG. M8398-F1 FOR DETAILS



▽ EL 15.00m
€ NEW eJV COMMScope RRZZVVT4S4-65D-R8-EC
PANEL ANTENNAS (3 OFF)

▽ EL 12.20m
€ NEW VODAFONE NOKIA RRU AKQJ (NR36) (3 OFF)

▽ EL 11.90m
€ NEW OPTUS ERICSSON RRU8863 (NR35) (3 OFF)

▽ EL 9.95m
€ NEW VODAFONE NOKIA AHEGA (L18/L21) (3 OFF)
€ NEW VODAFONE NOKIA RRU AHCA (L8.5) (3 OFF)
€ NEW VODAFONE NOKIA RRU AHPB (NR7) (3 OFF)
€ NEW OPTUS ERICSSON RRU8863 (LNR23) (2 OFF)
€ RESERVE OPTUS FUTURE RRU's (3 OFF)

▽ EL 9.10m
€ NEW OPTUS ERICSSON RRU4480 (L7/LN9) (3 OFF)
€ NEW OPTUS ERICSSON RRU4466 (L18/LNR21/L26) (3 OFF)
€ NEW VODAFONE NOKIA AHEGA (L18/L21) (3 OFF)
€ NEW OPTUS ERICSSON RRU8863 (LNR23) (1 OFF)

EXISTING OPTUS 450mm WIDE
CABLE TRAY UP TO UPPER ROOF

EXISTING UPPER ROOF ACCESS
LADDER (LAD-SAF TO BE
RE-CERTIFIED BY INDARA)

▽ EL 8.65m
TOP OF EXISTING PARAPET

▽ EL 8.29m
TOP OF EXISTING UPPER ROOF

▽ EL 5.94m
TOP OF EXISTING PARAPET

▽ EL 5.14m
TOP OF EXISTING UPPER ROOF

REUSE EXISTING SHARED H&S 6/12 HYBRID CABLE TO BE
ALLOCATED TO VODAFONE (1 OFF PER SECTOR, 3 OFF TOTAL)
(LENGTH: 35m). INSTALL NEW H&S 6/12 (10mm²) HYBRID CABLES
(1 OFF PER SECTOR, 3 OFF TOTAL) (LENGTH: 40m) FOR OPTUS
RUN IN EXISTING OPTUS 450mm WIDE CABLE LADDER. REMOVE
EXISTING AVA5-50 CABLES (12 OFF) FROM 450mm CABLE LADDER

▽ EL 0.00m (RL 168.10m AHD)
GROUND LEVEL

NORTHERN ELEVATION

SCALE 1:125

Rev	Date	Revision Details	Consultant	CAD	Designer	Verifier	Approver
A	02.04.25	ISSUED FOR CONSTRUCTION (5G UPGRADE (eJV-OH))	ALL TELECOS	TE	BC	AQ	HR
AB	17.06.15	AS BUILT ISSUED	VPL	RB	MA	RB	NR
A	31.10.14	ISSUED FOR CONSTRUCTION	VPL	PLB	MA	SM	DP
AB	23.11.11	AS BUILT ISSUED	AURECON	SB	NP	SZ	DP

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Client:

OPTUS

Project:

MOBILE NETWORK
AUSTRALIA
SITE No:- M8398
DOREEN TOWNSHIP-O
95 HAZEL GLEN DRIVE

Drawing Title:

SITE ELEVATION

Drawing Status:

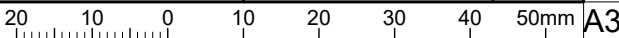
FOR CONSTRUCTION

Drawing No.

M8398-G4

Revision

A



A1 - PHYSICAL ANTENNA GROUP 1 & 2 DETAILS

ANTENNA DETAILS								
PHYSICAL ANTENNA GROUP (N2)		1						
CO-ORDINATES (N5)	LATITUDE	-37.58688						
	LONGITUDE	145.12794						
ANTENNA OPERATOR		JOINT						
PHYSICAL ANTENNA NUMBER (N3)		1						
PHYSICAL ANTENNA ID (N4)		11-J						
ANTENNA ALIAS		28-PORT TALL BF						
ANTENNA STATUS		NEW						
PHYSICAL AZIMUTH (°) (N6)		30						
PHYSICAL ANT CL EL (m) (N7)		15						
MECH DOWN-TILT (°)		0						
MODEL NO.		RRZZVVT4S4-65D-R8-EC						
MANUFACTURER		COMMS						
TYPE		PASSIVE						
NO. OF RF PORTS / TxRx		28						
PORT TYPE		4.3-10 / MLOC						
BAND CAPABILITY		07 / 09 / 18 / 21 / 23 / 26 / 23 / 35 / 36						
DIMENSIONS (HxWxD) (mm)		2688 × 498 × 197						
WEIGHT (kg)		62						
RF SECTOR & BAND ALLOCATION								
RF SECTOR ID / SUB-SECTOR ID		1						
PHYSICAL ARRAY ID	L11	L21	H12	H22	H13	H23	H14	H15
LOGICAL ANTENNA PORT	LP1 & LP2	LP3 & LP4	LP5 & LP6	LP7 & LP8	LP9 & LP10	LP11 & LP12	LP13 to LP20	LP21 to LP28
PHYSICAL ANTENNA PORT (OEM LABEL)	3 & 4	1 & 2	17 & 18	5 & 6	19 & 20	7 & 8	9 to 16	21 to 28
ETILT (N8)	2	2	2	2	2	2	2	2
ARRAY BEAM OFFSET (°)		0						
ARRAY BEAM AZIMUTH (°)		30						
ARRAY BAND CAPABILITY	7,8,9	7,8,9	18,21,23,26	18,21,23,26	18,21,23,26	18,21,23,26	23	35,36
ASSIGNED ARRAY OPERATOR	V	O	V	V	O	O	O	J
ASSIGNED RF PORT FREQUENCY 1	07	07/09	18/21	18/21	18/21/26	18/21/26	23	35
ASSIGNED RF PORT FREQUENCY 2	85							36
ASSIGNED RF PORT FREQUENCY 3								
ASSIGNED RF PORT FREQUENCY 4								

2							
-37.58688							
145.12794							
JOINT							
1							
21-J							
28-PORT TALL BF							
NEW							
150							
15							
0							
RRZZVVT4S4-65D-R8-EC							
COMMS							
PASSIVE							
28							
4.3-10 / MLOC							
07 / 09 / 18 / 21 / 23 / 26 / 23 / 35 / 36							
2688 × 498 × 197							
62							
2							
L11	L21	H12	H22	H13	H23	H14	H15
LP1 & LP2	LP3 & LP4	LP5 & LP6	LP7 & LP8	LP9 & LP10	LP11 & LP12	LP13 to LP20	LP21 to LP28
3 & 4	1 & 2	17 & 18	5 & 6	19 & 20	7 & 8	9 to 16	21 to 28
2	2	2	2	2	2	2	2
0							
150							
7,8,9	7,8,9	18,21,23,26	18,21,23,26	18,21,23,26	18,21,23,26	23	35,36
V	O	V	V	O	O	O	J
07	07/09	18/21	18/21	18/21/26	18/21/26	23	35
85							36

eJV ANTENNA SYSTEM CONFIGURATION FOR SECTOR 1 & 2

NOTES:

1. THIS ANTENNA TABLE SHALL BE READ IN CONJUNCTION WITH THE DETAIL RF PLUMBING DRAWING FOR THIS SITE.
2. A PHYSICAL SECTOR GROUP IS DEFINED AS A GROUP OF ANTENNAS GENERALLY ORIENTATED WITH SIMILAR AZIMUTHS AND EACH GROUP IS ASSIGNED A NUMBER 1, 2, 3, 4, ETC. THE PHYSICAL SECTOR GROUP MAY INCLUDE SINGLE BEAM OR MULTI-BEAM ANTENNAS. IN THE CASE OF MULTI-BEAM ANTENNAS THE RF SECTOR MAY DIFFER TO THE PHYSICAL SECTOR.
3. THE PHYSICAL ANTENNA NUMBER IS DEFINED BY ASSIGNING SEQUENCED NUMBERS TO EACH JV OPERATOR'S ANTENNAS WITHIN A PHYSICAL SECTOR GROUP, STARTING FROM LEFT TO RIGHT, TOP TO BOTTOM.
4. THE PHYSICAL ANTENNA ID IS DEFINED BY THE NUMBER MADE UP BY CONCATENATING THE PHYSICAL SECTOR GROUP NUMBER WITH THE PHYSICAL ANTENNA NUMBER & THE FIRST LETTER FROM THE OPERATOR'S NAME OR JOINT (O, V, J), SAME AS THE SAO NUMBER, AS SPECIFIED IN OSD-070 SECTION 5.6.
5. ANTENNA CO-ORDINATES ARE TO BE IN GDA94/GDA2020 REFERENCE SYSTEM, SPECIFIED FOR CENTRE OF TOWER TO NEAREST METRE.
6. THE PHYSICAL AZIMUTH (°) IS DEFINED AS THE DIRECTION OF THE PHYSICAL ANTENNA BORE SIGHT WITH RESPECT TO TRUE NORTH. AZIMUTHS OF EACH BEAM OF MULTI-BEAM ANTENNAS ARE NOT DOCUMENTED IN THIS TABLE.
7. THE PHYSICAL ANT CL EL (m) IS DEFINED AS THE MID-HEIGHT OF THE PHYSICAL ANTENNA. THE HEIGHT OF INDIVIDUAL BEAMS OR DIFFERENT SEGMENTS OF MULTI-BEAM OR HYBRID ANTENNAS (IPAA) ARE NOT DOCUMENTED IN THIS TABLE.
8. ELECTRICAL DOWN-TILTS ARE ONLY ACCURATE AT TIME OF BUILD AS THEY ARE FREQUENTLY OPTIMISED. ALWAYS REFER TO CURRENT VALUES IN MNIS.

[illegible]

A2 - PHYSICAL ANTENNA GROUP 3 DETAILS

ANTENNA DETAILS								
PHYSICAL ANTENNA GROUP (N2)		3						
CO-ORDINATES (N5)	LATITUDE	-37.58688						
	LONGITUDE	145.12794						
ANTENNA OPERATOR		JOINT						
PHYSICAL ANTENNA NUMBER (N3)		1						
PHYSICAL ANTENNA ID (N4)		31-J						
ANTENNA ALIAS		28-PORT TALL BF						
ANTENNA STATUS		NEW						
PHYSICAL AZIMUTH (°) (N6)		260						
PHYSICAL ANT CL EL (m) (N7)		15						
MECH DOWN-TILT (°)		0						
MODEL NO.		RRZZVVT4S4-65D-R8-EC						
MANUFACTURER		COMMS						
TYPE		PASSIVE						
NO. OF RF PORTS / TxRx		28						
PORT TYPE		4.3-10 / MLOC						
BAND CAPABILITY		07 / 09 / 18 / 21 / 23 / 26 / 23 / 35 / 36						
DIMENSIONS (HxWxD) (mm)		2688 × 498 × 197						
WEIGHT (kg)		62						
RF SECTOR & BAND ALLOCATION								
RF SECTOR ID / SUB-SECTOR ID		3						
PHYSICAL ARRAY ID	L11	L21	H12	H22	H13	H23	H14	H15
LOGICAL ANTENNA PORT	LP1 & LP2	LP3 & LP4	LP5 & LP6	LP7 & LP8	LP9 & LP10	LP11 & LP12	LP13 to LP20	LP21 to LP28
PHYSICAL ANTENNA PORT (OEM LABEL)	3 & 4	1 & 2	17 & 18	5 & 6	19 & 20	7 & 8	9 to 16	21 to 28
ETILT (N8)	2	2	2	2	2	2	2	2
ARRAY BEAM OFFSET (°)		0						
ARRAY BEAM AZIMUTH (°)		260						
ARRAY BAND CAPABILITY	7,8,9	7,8,9	18,21,23,26	18,21,23,26	18,21,23,26	18,21,23,26	23	35,36
ASSIGNED ARRAY OPERATOR	V	O	V	V	O	O	O	J
ASSIGNED RF PORT FREQUENCY 1	07	07/09	18/21	18/21	18/21/26	18/21/26	23	35
ASSIGNED RF PORT FREQUENCY 2	85							36
ASSIGNED RF PORT FREQUENCY 3								
ASSIGNED RF PORT FREQUENCY 4								

eJV ANTENNA SYSTEM CONFIGURATION FOR SECTOR 3

										 ServiceStream				Client: 				Project: MOBILE NETWORK AUSTRALIA SITE No:- M8398 DOREEN TOWNSHIP-O 95 HAZEL GLEN DRIVE				Drawing Title: eJV ANTENNA SYSTEM CONFIGURATION - SHEET 2 OF 2							
A 02.04.25 ISSUED FOR CONSTRUCTION (5G UPGRADE (eJV-OH)) ALL TELECOS TE BC AQ HR														Drawing Status: FOR CONSTRUCTION				Drawing No. M8398-A2				Revision A							
Rev Date Revision Details Consultant CAD Designer Verifier Approver																													

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50mm

A3

PHYSICAL ASSET SUMMARY TABLE

ANCILLARY DEVICES											
EQUIPMENT ALIAS	PART NO.	STATUS	LOCATION	OPERATOR	PHY ANT GRP	QTY	FREQUENCY	MFR	RF PORT TYPE	DIM (H x W x D)	WT (Kg)
850 REJ	E14V00P21	NEW	ANTENNA	O	1	1	85	COMMS	4.3-10	257 × 179.5 × 124	5.4
850 REJ	E14V00P21	NEW	ANTENNA	O	2	1	85	COMMS	4.3-10	257 × 179.5 × 124	5.4
850 REJ	E14V00P21	NEW	ANTENNA	O	3	1	85	COMMS	4.3-10	257 × 179.5 × 124	5.4
COM2	E14F60P18	NEW	ANTENNA	V	1	1	07 / 85 / 09	COMMS	4.3-10	272 × 254 × 111	8.5
COM2	E14F60P18	NEW	ANTENNA	V	2	1	07 / 85 / 09	COMMS	4.3-10	272 × 254 × 111	8.5
COM2	E14F60P18	NEW	ANTENNA	V	3	1	07 / 85 / 09	COMMS	4.3-10	272 × 254 × 111	8.5
COM20d	E14F07P16	NEW	ANTENNA	J	1	1	35	COMMS	4.3-10	176 × 439.5 × 58.5	8.2
COM20d	E14F07P16	NEW	ANTENNA	J	2	1	35	COMMS	4.3-10	176 × 439.5 × 58.5	8.2
COM20d	E14F07P16	NEW	ANTENNA	J	3	1	35	COMMS	4.3-10	176 × 439.5 × 58.5	8.2

RADIO UNITS												
EQUIPMENT ALIAS	PART NO.	STATUS	LOCATION	OPERATOR	PHY ANT GRP	QTY	RF PORT CONNECTED	MFR	NO. RF PORTS / TxRx	RF PORT TYPE	DIM (H x W x D)	WT (Kg)
RRU 700/900	RRU4480 B28/B8	NEW	ANTENNA	O	1	1	2	ERICSSON	4T4R	4.3-10	588 × 398 × 195	38.1
RRU 1800/2100/2600	RRU4466 B3/B1/B7	NEW	ANTENNA	O	1	1	4	ERICSSON	4T4R	4.3-10	666 × 401 × 201	45
RRU 2300	RRU8863 B40	NEW	ANTENNA	O	1	1	8	ERICSSON	8T8R	Nex-10	478 × 375 × 155	24
RRU 3500	RRU8863 B42	NEW	ANTENNA	O	1	1	8	ERICSSON	8T8R	Nex-10	478 × 375 × 155	24
RRU 700/900	RRU4480 B28/B8	NEW	ANTENNA	O	2	1	2	ERICSSON	4T4R	4.3-10	588 × 398 × 195	38.1
RRU 1800/2100/2600	RRU4466 B3/B1/B7	NEW	ANTENNA	O	2	1	4	ERICSSON	4T4R	4.3-10	666 × 401 × 201	45
RRU 2300	RRU8863 B40	NEW	ANTENNA	O	2	1	8	ERICSSON	8T8R	Nex-10	478 × 375 × 155	24
RRU 3500	RRU8863 B42	NEW	ANTENNA	O	2	1	8	ERICSSON	8T8R	Nex-10	478 × 375 × 155	24
RRU 700/900	RRU4480 B28/B8	NEW	ANTENNA	O	3	1	2	ERICSSON	4T4R	4.3-10	588 × 398 × 195	38.1
RRU 1800/2100/2600	RRU4466 B3/B1/B7	NEW	ANTENNA	O	3	1	4	ERICSSON	4T4R	4.3-10	666 × 401 × 201	45
RRU 2300	RRU8863 B40	NEW	ANTENNA	O	3	1	8	ERICSSON	8T8R	Nex-10	478 × 375 × 155	24
RRU 3500	RRU8863 B42	NEW	ANTENNA	O	3	1	8	ERICSSON	8T8R	Nex-10	478 × 375 × 155	24
RRU 700	AHPB B28	NEW	ANTENNA	V	1	1	2	NOKIA	4T4R	4.3-10	630 × 321 × 142	25.259
RRU 850	AHCA B5	NEW	ANTENNA	V	1	1	2	NOKIA	4T4R	4.3-10	427.5 × 309 × 189	17.588
RRU 1800/2100	AHEGA B3/B1	NEW	ANTENNA	V	1	2	4	NOKIA	2T4R	4.3-10	630 × 321 × 128.5	24.888
RRU 3500	AKQJ N78	NEW	ANTENNA	V	1	1	8	NOKIA	8	4.3-10	445 × 360 × 110	16
RRU 700	AHPB B28	NEW	ANTENNA	V	2	1	2	NOKIA	4T4R	4.3-10	630 × 321 × 142	25.259
RRU 850	AHCA B5	NEW	ANTENNA	V	2	1	2	NOKIA	4T4R	4.3-10	427.5 × 309 × 189	17.588
RRU 1800/2100	AHEGA B3/B1	NEW	ANTENNA	V	2	2	4	NOKIA	2T4R	4.3-10	630 × 321 × 128.5	24.888
RRU 3500	AKQJ N78	NEW	ANTENNA	V	2	1	8	NOKIA	8	4.3-10	445 × 360 × 110	16
RRU 700	AHPB B28	NEW	ANTENNA	V	3	1	2	NOKIA	4T4R	4.3-10	630 × 321 × 142	25.259
RRU 850	AHCA B5	NEW	ANTENNA	V	3	1	2	NOKIA	4T4R	4.3-10	427.5 × 309 × 189	17.588
RRU 1800/2100	AHEGA B3/B1	NEW	ANTENNA	V	3	2	4	NOKIA	2T4R	4.3-10	630 × 321 × 128.5	24.888
RRU 3500	AKQJ N78	NEW	ANTENNA	V	3	1	8	NOKIA	8	4.3-10	445 × 360 × 110	16
FUTURE-RRU	ERICSSON	FUTURE	ANTENNA	O	1	1	-	-	-	-	588 × 398 × 195	38.1
FUTURE-RRU	ERICSSON	FUTURE	ANTENNA	O	2	1	-	-	-	-	588 × 398 × 195	38.1
FUTURE-RRU	ERICSSON	FUTURE	ANTENNA	O	3	1	-	-	-	-	588 × 398 × 195	38.1

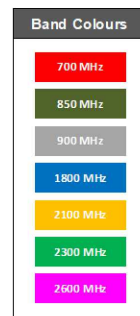
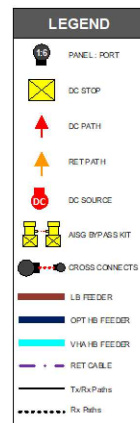
COAXIAL FEEDER / HFDC TRUNK CABLES								
CABLE ALIAS	STATUS	LENGTH (m)	OPERATOR	PHY ANT GRP	QTY	COAX BAND ASIGNMENT	MFR	
HFDC 6/12 C10	NEW	40	O	1	1	LBHB	H+S	
HFDC 6/12 C10	NEW	40	O	2	1	LBHB	H+S	
HFDC 6/12 C10	NEW	40	O	3	1	LBHB	H+S	
HFDC 6/12 C6	EXISTING	35	V	1	1	LBHB	H+S	
HFDC 6/12 C6	EXISTING	35	V	2	1	LBHB	H+S	
HFDC 6/12 C6	EXISTING	35	V	3	1	LBHB	H+S	

NOTES:

1. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH RF PLUMBING DIAGRAM DRG. M8398-P1 & P2.
2. INFORMATION IN THE TABLES SUPPLIED AND VERIFIED BY OPTUS.
3. ANCILLARIES REFER TO ITEMS AT OR NEAR THE ANTENNA.
4. TRUNK LENGTHS ARE ESTIMATED, ROUNDED UP TO THE NEXT 10m.
5. RF TAIL LENGTH FOR 5G RRU_s RANGES IS 5m MAX.
6. RF TAIL LENGTH FOR 4G RRU_s RANGES FROM 9m TO 11m. REFER TABLE

TAIL LENGTH	
RRU4466	8m
RRU4480	10m
AHEGA	9m
AHPB	11m
AHCA	10m

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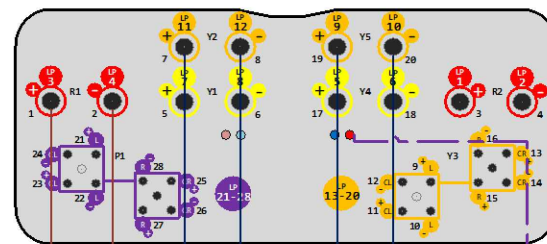
Optus Detailed Plumbing
Sector 1,2,3

TPG DPD follow EBP 10.4

Sector 1: eJV RSD7.SB1
Sector 2: eJV RSD7.SB1
Sector 3: eJV RSD7.SB1

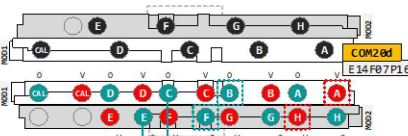
NOTES

N26G Design only

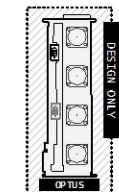
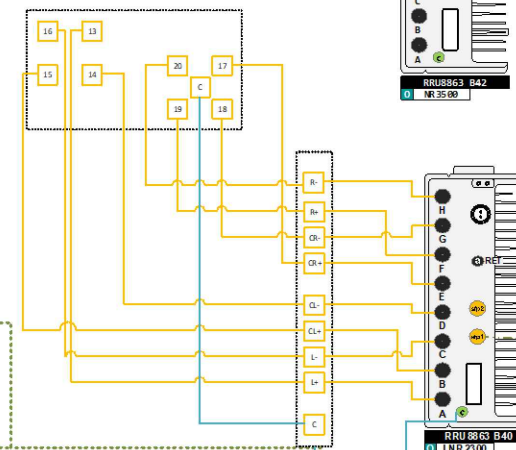
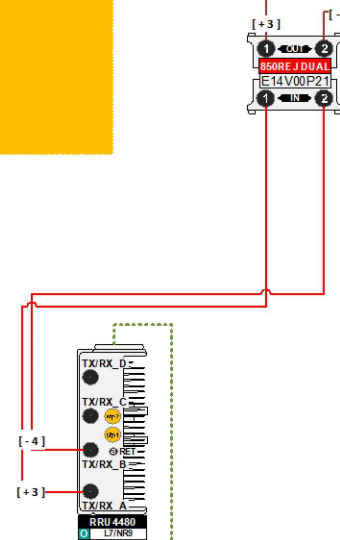


OPERATOR OPTUS
MODEL RRZVVT4S4-650-RB-EC

● AISG1 IN ● AISG2 IN
● AISG1 OUT ● AISG2 OUT



NR3500 RRU TO COM20b,c,d MAPPING TABLE					
TPG AZQJ	OPTUS AZQJ / ARQJ	OPTUS RRU863	COM20b,c,d	ANT	
8	8	H	A	R+	
4	4	F	B	R-	
7	7	G	C	CR+	
3	3	E	D	CR-	
5	5	C	E	L+	
1	1	A	F	L-	
6	6	D	G	CL+	
2	2	B	H	CL-	
CAL	CAL	CAL	CAL	CAL	



OPERATOR OPTUS
GPS ERICSSON

"For BBU connection instructions, please refer to the relevant IME documents".

OPTUS RF PLUMBING DIAGRAM

PER SECTOR

NOTE:

- THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRAWING M8398-A1, A2 AND A3.

Rev	Date	Revision Details	Consultant	CAD	Designer	Verifier	Approver
A	02.04.25	ISSUED FOR CONSTRUCTION (5G UPGRADE (eJV-OH))	ALL TELECOMS	TE	BC	AQ	HR
AB	17.06.15	AS BUILT ISSUED	VPL	RB	MA	RB	NR
A	31.10.14	ISSUED FOR CONSTRUCTION	VPL	PLB	MA	SM	DP

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Client:

OPTUS

Project:

MOBILE NETWORK
AUSTRALIA
SITE No:- M8398
DOREEN TOWNSHIP-O
95 HAZEL GLEN DRIVE

Drawing Title:

OPTUS RF PLUMBING DIAGRAM

Drawing Status:

FOR CONSTRUCTION

Drawing No.

M8398-P1

Revision

A

20 10 0 10 20 30 40 50mm A3

LEGEND

PANEL PORT

DC STOP

DC PATH

RET PATH

DC SOURCE

AISG BYPASS KIT

CROSS CONNECTS

LB FEEDER

OPT HB FEEDER

VIA HB FEEDER

RET CABLE

Tx/Rx Paths

Rx Paths

BAND COLOURS

700 MHz

850 MHz

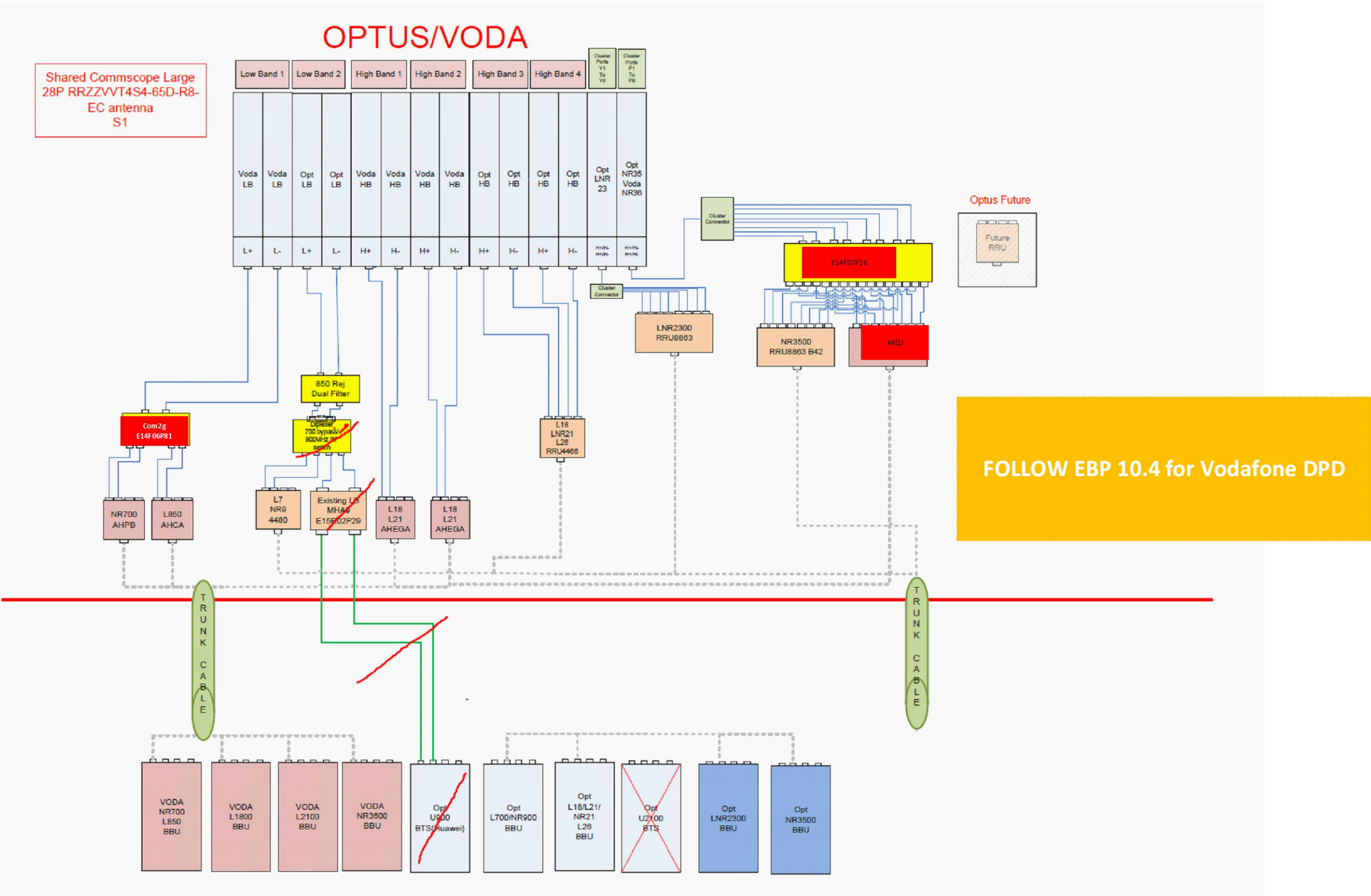
900 MHz

1800 MHz

2100 MHz

2300 MHz

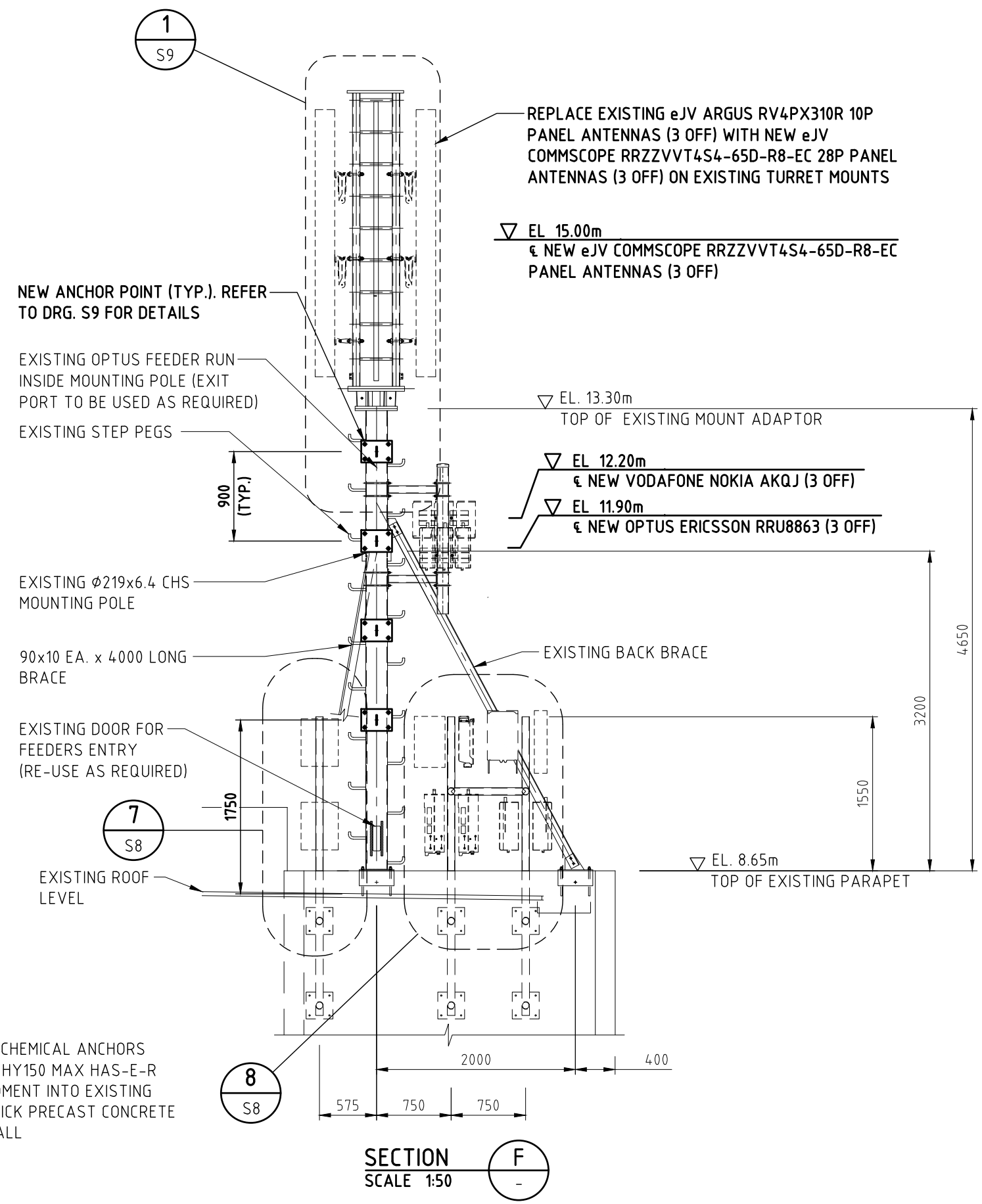
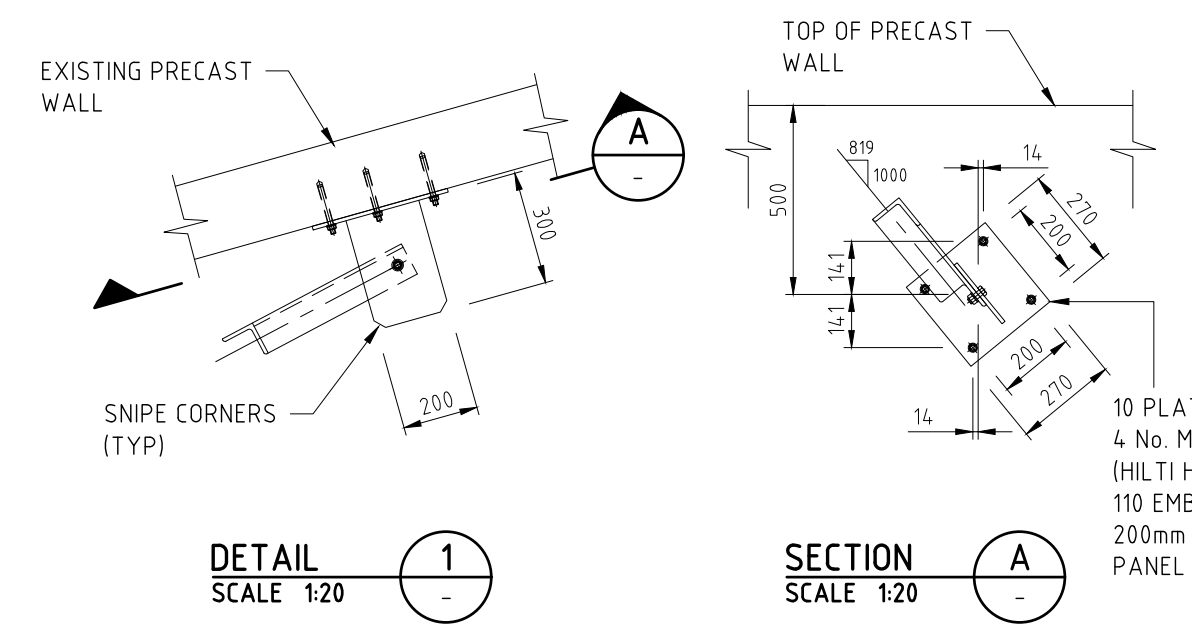
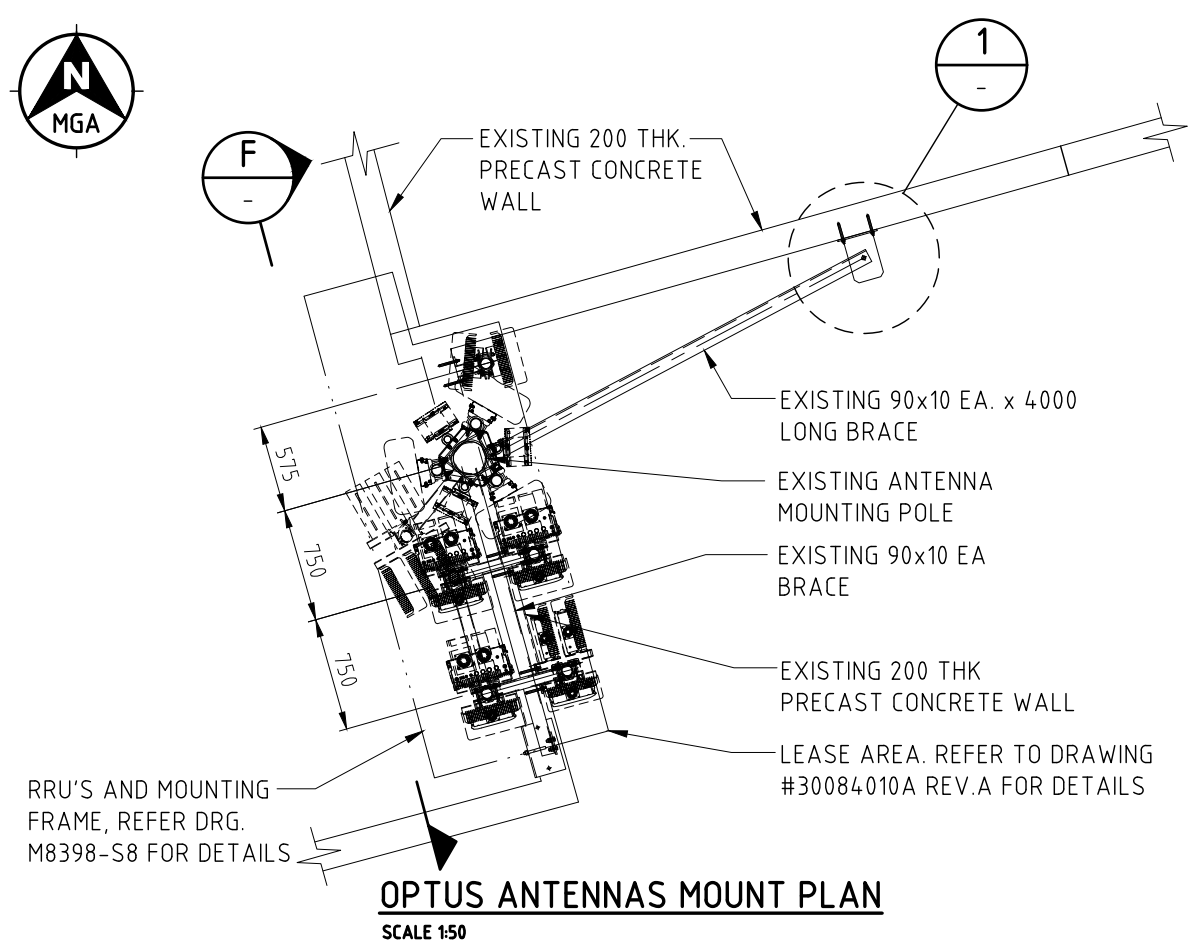
2600 MHz



NOTE:
1. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH
DRAWING M8398-A1, A2 AND A3.

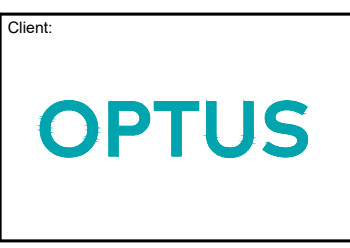
OPTUS & VODAFONE RF PLUMBING DIAGRAM
PER SECTOR

										Client: MOBILE NETWORK AUSTRALIA SITE No:- M8398 DOREEN TOWNSHIP-O 95 HAZEL GLEN DRIVE										Drawing Title: OPTUS & VODAFONE RF PLUMBING DIAGRAM Drawing Status: FOR CONSTRUCTION										Drawing No. M8398-P2 Revision A									
Rev: A Date: 02.04.25 Revision Details: ISSUED FOR CONSTRUCTION (5G UPGRADE (eJV-OH))										Consultant: ALL TELECOMS CAD: TE Designer: BC Verifier: AQ Approver: HR										COPYRIGHT © OPTUS MOBILE PTY LTD ABN 65 054 365 696. ALL RIGHTS RESERVED. VERSION 2.0 NOVEMBER 2013										20 10 0 10 20 30 40 50mm A3									



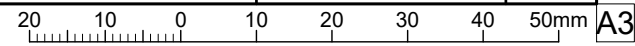
Rev	Date	Revision Details	Consultant	CAD	Designer	Verifier	Approver
A	02.04.25	ISSUED FOR CONSTRUCTION (5G UPGRADE (eJV-OHI))	ALL TELECOMS	TE	BC	AQ	HR
AB	17.06.15	AS BUILT ISSUED	VPL	RB	MA	RB	NR
A	31.10.14	ISSUED FOR CONSTRUCTION	VPL	PLB	MA	SM	DP
AB	23.11.11	AS BUILT ISSUED	AURECON	SB	NP	SZ	DP
B	31.05.11	STRUCTURAL DETAILS AMENDED	AURECON	JND	JL	SZ	DP
A	09.12.10	ISSUED FOR CONSTRUCTION	AURECON	JND	JL	SZ	DP

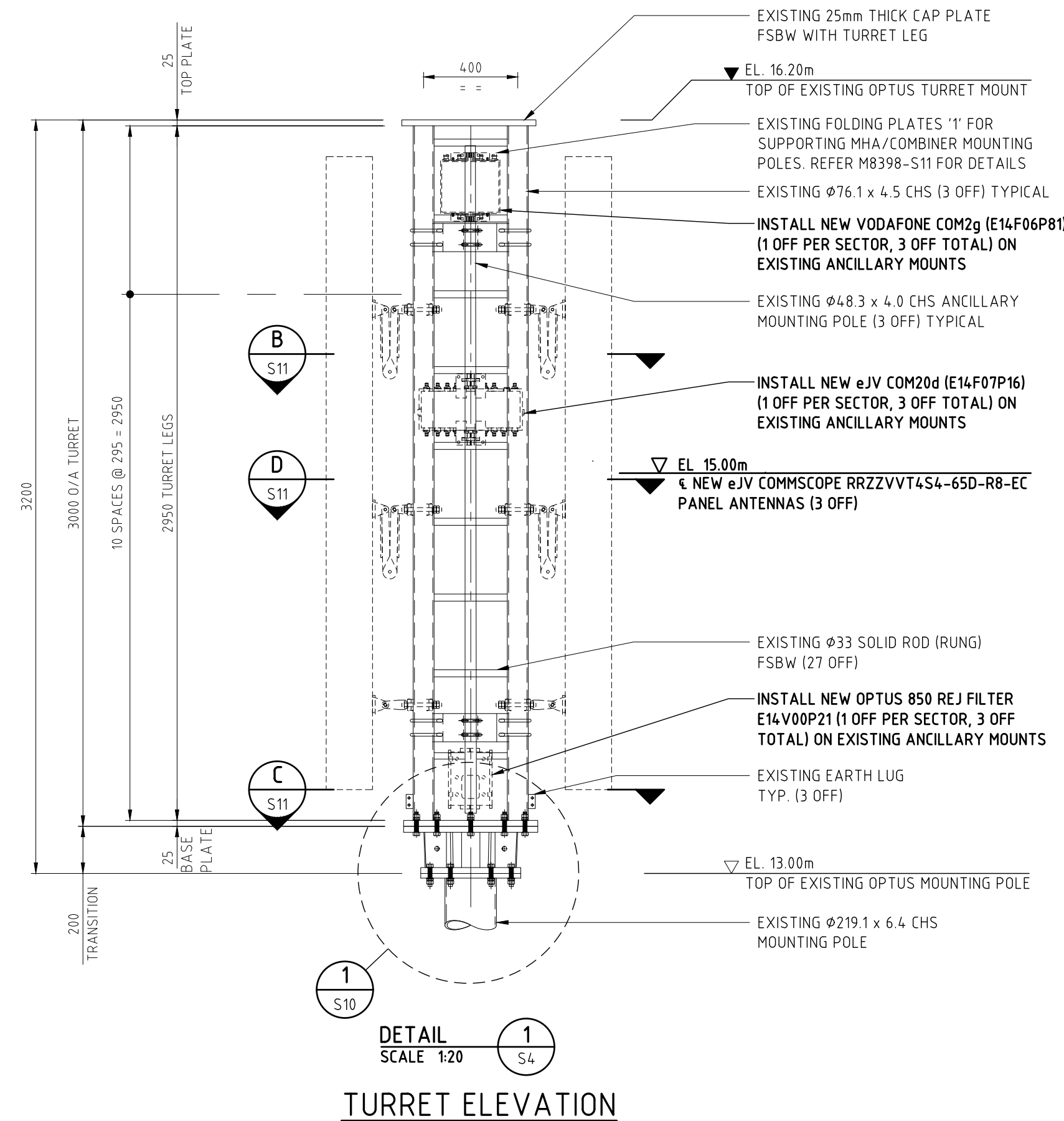
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Client:
Project: MOBILE NETWORK AUSTRALIA
SITE No:- M8398
DOREEN TOWNSHIP-O
95 HAZEL GLEN DRIVE

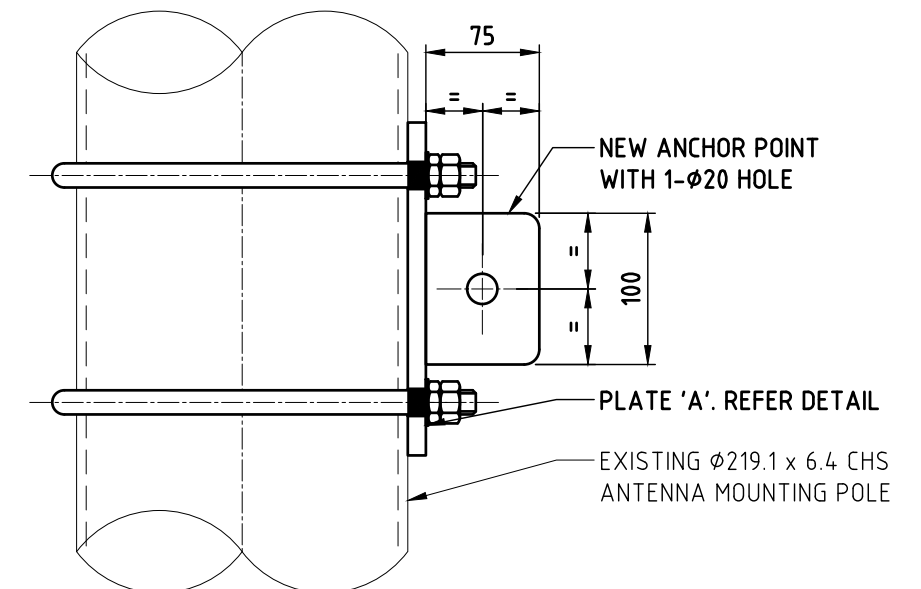
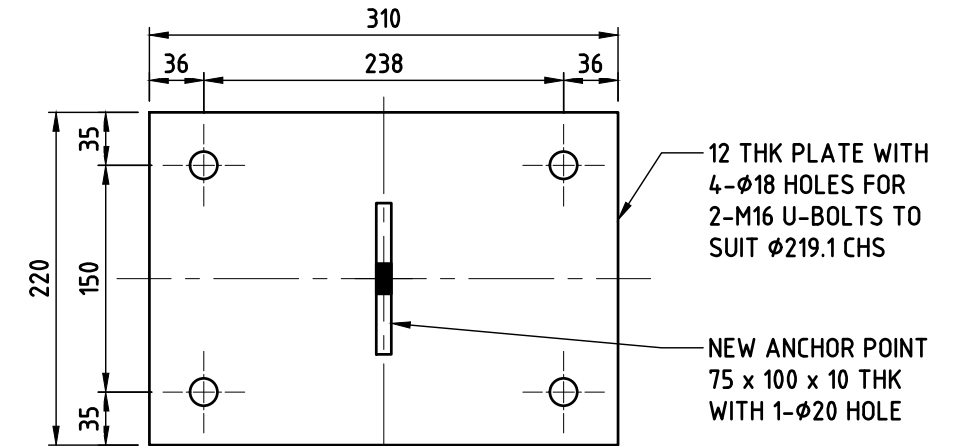
Drawing Title:
ANTENNAS MOUNT DETAILS SHEET 1
Drawing Status:
FOR CONSTRUCTION
Drawing No.
M8398-S4
Revision
A





NOTES:

1. REFER TO OPTUS STANDARD DRG. OSD-100 FOR CONSTRUCTION NOTES.
2. ALL WELDS TO BE 6mm FILLET WELD ALL ROUND UNLESS NOTED.
3. UNLESS OTHERWISE SPECIFIED, ALL BOLTS SHALL BE GRADE AND STRENGTH 8.8/S AND SHALL HAVE LOCK NUTS FITTED.
4. ALL STEELWORK TO BE PRIOR ASSEMBLED IN WORKSHOP PRIOR TO HOT DIP GALVANISING.
5. NO SET-DOWN OR STOCKPILING OF EQUIPMENT IS PERMITTED DIRECTLY ON THE STEEL ROOF SHEETING. IF MATERIAL SET-DOWN IS REQUIRED, THE SHEETING SHALL BE PROTECTED AND THE LOCATION SHALL BE SPECIFIED BY A STRUCTURAL ENGINEER.



Rev	Date	Revision Details	Consultant	CAD	Designer	Verifier	Approver
A	02.04.25	ISSUED FOR CONSTRUCTION (5G UPGRADE (eJV-OHI))	ALL TELECOMS	TE	BC	AQ	HR
AB	17.06.15	AS BUILT ISSUED	VPL	RB	MA	RB	NR
A	31.10.14	ISSUED FOR CONSTRUCTION	VPL	PLB	KS	LC	DP

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Client:

OPTUS

Project:

MOBILE NETWORK AUSTRALIA
SITE No:- M8398
DOREEN TOWNSHIP-O
95 HAZEL GLEN DRIVE

Drawing Title:

ANTENNA TURRET MOUNT
DETAIL SHEET 1 OF 3

Drawing Status:

FOR CONSTRUCTION

Drawing No.

M8398-S9

Revision

A

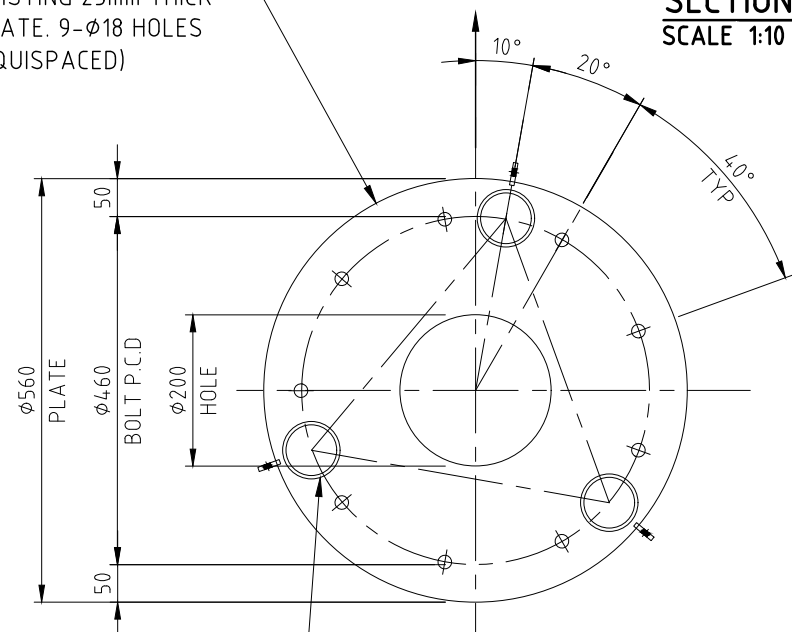
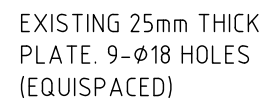
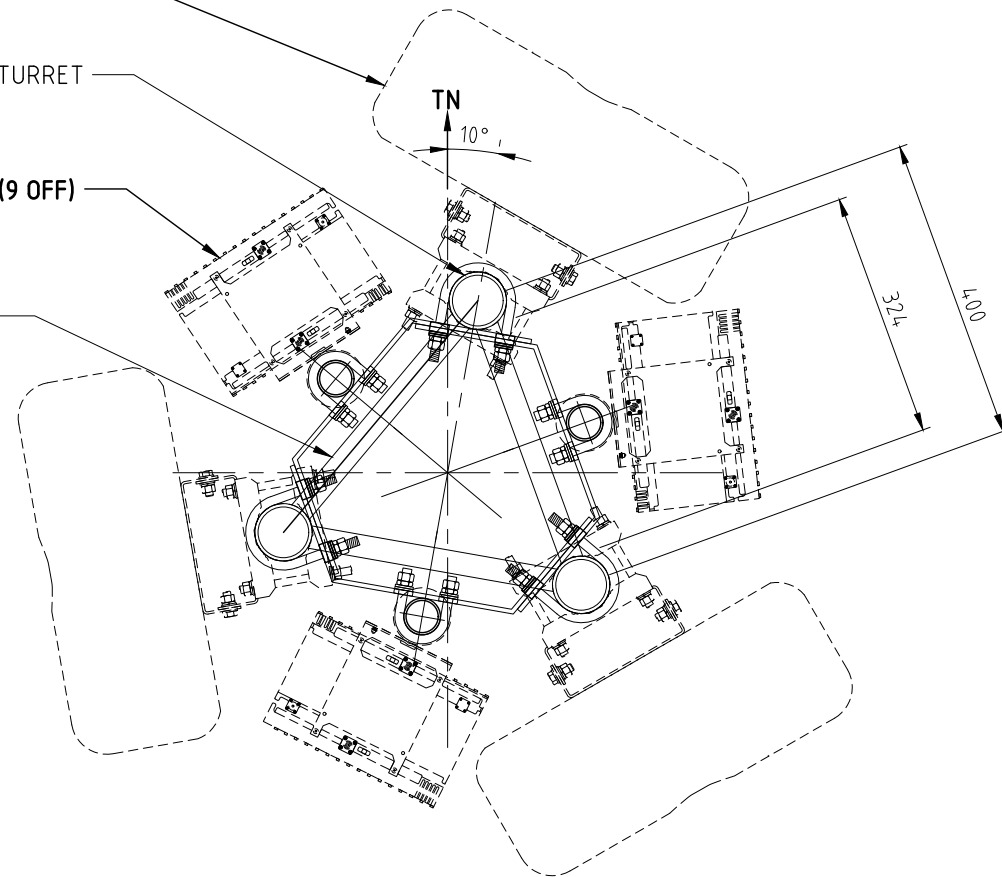
20 10 0 10 20 30 40 50mm **A3**



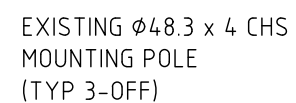
EXISTING $\phi 76.1 \times 4.5$ CHS TURRET
LEGS (3 OFF)

NEW OPTUS ANCILLARIES (9 OFF)

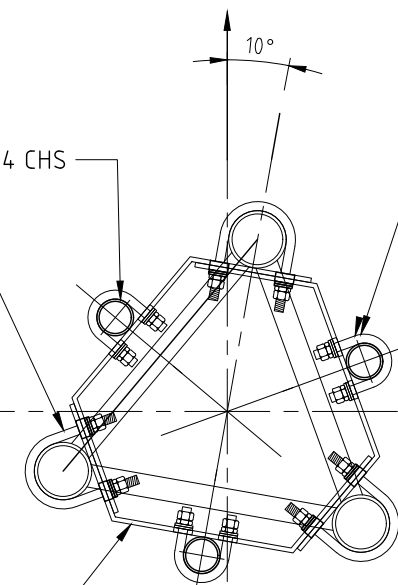
EXISTING Ø33 SOLID ROD
RUNGS FSBW TO CHS
TURRET LEGS TYP.



SECTION
SCALE 1:10

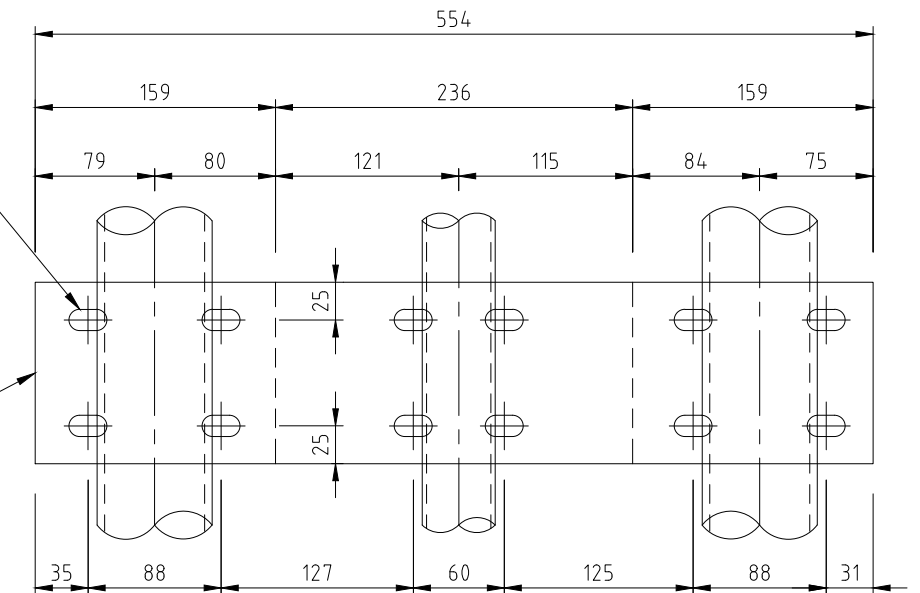
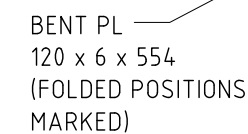
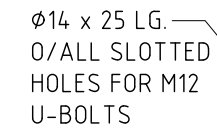


EXISTING M12
U-BOLTS



EXISTING FOLDED
PLATE '1' FOR
MHA/COMBINERS
MOUNTING POLES
(TYP. 3-OFF)

SECTION
SCALE 1:10

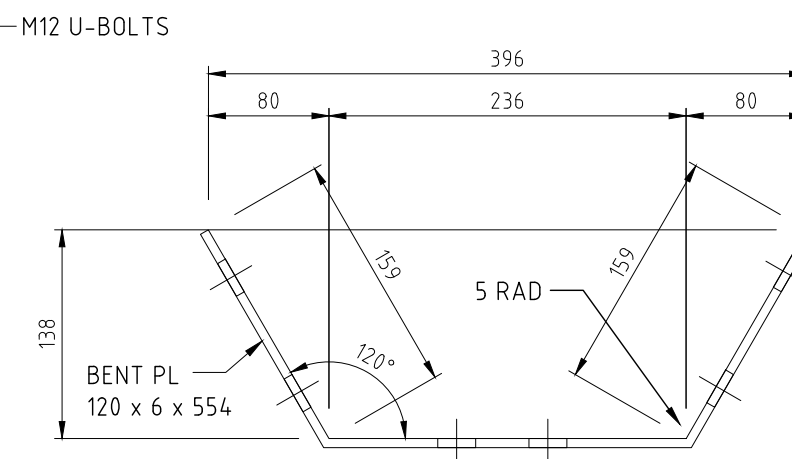


PRE-FOLDED PLATE '1'

SCALE 1:5

NOTES:

1. REFER TO OPTUS STANDARD DRG. OSD-100 FOR CONSTRUCTION NOTES.
2. ALL WELDS TO BE 6mm FILLET WELD ALL ROUND UNLESS NOTED.
3. UNLESS OTHERWISE SPECIFIED, ALL BOLTS SHALL BE GRADE AND STRENGTH 8.8/S AND SHALL HAVE LOCK NUTS FITTED.
4. ALL STEELWORK TO BE PRIOR ASSEMBLED IN WORKSHOP PRIOR TO HOT DIP GALVANISING.
5. ALL INFORMATION TO BE CHECKED ON SITE PRIOR TO FABRICATION & CONSTRUCTION.



DETAIL OF FOLDED PLATE '1'

SCALE 1:5

EXISTING Ø76.1 x 4.5 CHS
TURRET LEGS FSBW TO
BASE AND CAP PLATE

SECTION
SCALE 1:10

A	02.04.25	ISSUED FOR CONSTRUCTION (5G UPGRADE (e)JV-OH))	ALL TELECOS	TE	BC		AQ		HR
AB	17.06.15	AS BUILT ISSUED	VPL	RB	MA		RB		NR
A	31.10.14	ISSUED FOR CONSTRUCTION	VPL	PLB	KS		LC		DP
Rev	Date	Revision Details	Consultant	CAD	Designer		Verifier		Approver



Client:	
---------	--



Project:

Project: MOBILE NETWORK
AUSTRALIA
SITE No:- M8398
DOREEN TOWNSHIP-O
95 HAZEL GLEN DRIVE

Drawing Title:

ANTENNA TURRET MOUNT DETAIL SHEET 3 OF 3

Drawing Status:

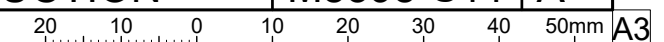
FOR CONSTRUCTION

Drawing No.

Drawing No.
M8398-S11

Revision

A



2900

RECOVER EXISTING VODAFONE L8.5
& LU/9 HUAWEI BTS RACK (1 OFF)

EXISTING GLAND
PLATE (600x400)

NEW ELTEK
37RU MER
HFDC RACK
FOR DCDU'S

**NEW OPTUS
37RU MER
BTS RACK
FIBRE AND 4G &
5G BBU**

**NEW VODAFONE
37RU MER
BTS RACK
FIBRE AND 4G &
5G BBU**

2400

NEW
VODAFONE
VERTIV
PSU-2

RECOVER EXISTING —
COMBINER RACK
(1 OFF) (ALL
COMBINERS AS WELL)

NEW OPTUS
VERTIV
PSU-1

REFER TO NOTES
7 & 8

EXISTING
TX MER
RACK

— EXISTING 600mm O/H
CABLE TRAY

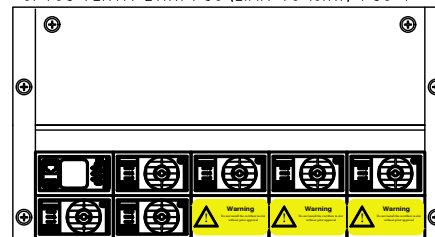
— EXISTING 150mm O/H
CABLE TRAY

EXISTING LENMIC
L600P
A/C UNITS (3 OFF)
AND LENMIC AC
ECO-FAN WITH DC
INVERTER
(1 OFF BELOW)

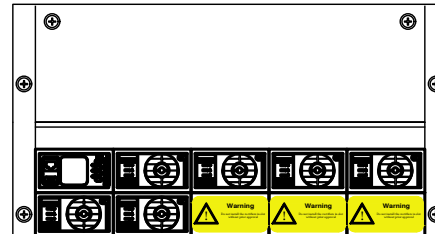
— EXISTING DISTRIBUTION BOARD WITH PLC & SPD. REFER TO NOTE 3

— EXISTING METER BOX/EXTERNAL DISTRIBUTION PANEL

PSU RECTIFIER RESTRICTION CONFIGURATION TABLE:
OPTUS VERTIV 27kW PSU (LIMIT TO 18kW): PSU-1



VODAFONE VERTIV 27kW PSU (LIMIT TO 18kW): PSU-2



7. ALL EXTERNAL ALARMS (ENVIRONMENTAL ETC) TO BE MIGRATED TO 4G AND 5G BBU.

8. ALL KRONE PANELS TO BE REPLACED WITH RJ45 PATCH PANEL.

SCALE 1:20

A	02.04.25	ISSUED FOR CONSTRUCTION (5G UPGRADE (eJV-OH))	ALL TELECOS	TE	BC		AQ		HR
AB	17.06.15	AS BUILT ISSUED	VPL	RB	MA		RB		NR
A	31.10.14	ISSUED FOR CONSTRUCTION	VPL	PLB	MA		SM		DP
Rev	Date	Revision Details	Consultant	CAD	Designer		Verifier		Approver



Client:	
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OPTUS

Project:

Project: MOBILE NETWORK
AUSTRALIA
SITE No:- M8398
DOREEN TOWNSHIP-O
95 HAZEL GLEN DRIVE

Drawing Title:

EQUIPMENT SHELTER LAYOUT PLAN

Drawing Status:

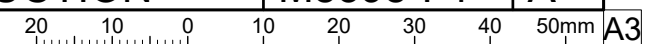
FOR CONSTRUCTION

Drawing No.

Drawing No.
M8398-F1

Revision

Revision
A



1. THE CONTRACTOR SHALL COMPLY WITH THE REQUIREMENTS OF THE SPECIFICATIONS FOR ALL WORKS AND REQUIREMENTS UNLESS NOTED OTHERWISE.
2. THE CONTRACTOR OR REPRESENTATIVE SHALL BE RESPONSIBLE FOR LIAISONS WITH THE PROPERTY OWNER REGARDING CONSTRUCTION OF THE INSTALLATION. THE PROPERTY MUST REMAIN SERVICEABLE AND OPERATIONAL AT ALL TIMES UNLESS AGREED WITH THE PROPERTY OWNER.
3. THE CONTRACTOR SHALL CONFIRM ALL DIMENSIONS ON SITE PRIOR TO FABRICATION AND CONSTRUCTION.
4. REFER TO GIVEN DIMENSIONS ONLY, DRAWINGS SHOULD NOT TO BE SCALED. DIMENSIONS ARE IN MILLIMETRES UNO.
5. SPECIFIED PRODUCTS (OR THEIR APPROVED EQUIVALENTS) SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S SPECIFICATIONS.
6. THE CONTRACTOR SHALL USE DROP SHEETS OR ANY OTHER METHOD DEEMED NECESSARY TO PROTECT THE EXISTING BUILDING FROM DAMAGE DURING CONSTRUCTION.
7. NORTH POINT AS SHOWN ON DRAWINGS INDICATES MGA NORTH (GDA 94 DATUM) UNLESS NOTED OTHERWISE.
8. ANTENNA AZIMUTHS ARE SPECIFIED IN DEGREES REFERENCED TO TRUE NORTH (TN).
9. ALL LEVELS ARE EXPRESSED IN METRES TO AUSTRALIAN HEIGHT DATUM (AHD).

1. THE CONTRACTOR SHALL BE RESPONSIBLE FOR SET OUT OF THE SHELTER OR OUTDOOR UNITS IN ACCORDANCE WITH THE DESIGN INTENT AS SHOWN ON THE DRAWINGS.
2. THE CONTRACTOR SHALL BE RESPONSIBLE FOR SET OUT OF THE CABLE LADDER IN ACCORDANCE WITH THE DESIGN INTENT AS SHOWN ON THE DRAWINGS.
3. CONCRETE WORKS TOLERANCES SHALL COMPLY WITH CLAUSE 17.5 OF AS 3600.
4. STRUCTURAL STEELWORK TOLERANCES SHALL COMPLY WITH CLAUSE 15.3 OF AS 4100.

CONSTRUCTION DETAILS ON OPTUS STANDARD DRAWINGS SHALL BE ONLY BE USED WITHIN THE SPECIFIED DESIGN CRITERIA. THE OPTUS DESIGN CONSULTANT SHALL CONFIRM APPLICABILITY TO THE SITE. THE OPTUS CONTRACTOR SHALL CONFIRM ACTUAL SITE CONDITIONS ARE SAME AS STATED ON THE STANDARD DRAWINGS REFERENCED.

1. ALL MATERIALS AND WORKMANSHIP TO BE IN ACCORDANCE WITH AS 4100, AS 1657 AND AS/NZS 1554.1.
2. ALL STEELWORK SHALL BE IN ACCORDANCE WITH AS/NZS 3679.1 (GRADE 300) FOR HOT-ROLLED SECTIONS AND BARS), AS/NZS 3678 (GRADE 250) FOR HOT-ROLLED PLATES AND AS 1163 (GRADE 250 (MIN)) FOR HOLLOW SECTIONS.
3. THE CONTRACTOR SHALL CONFIRM ALL DIMENSIONS ON SITE PRIOR TO FABRICATION.
4. ALL PREFABRICATED STEELWORK SHALL BE HOT-DIP GALVANISED AFTER FABRICATION, IN ACCORDANCE WITH AS/NZS 4680. PLUG AND SEAL WATERTIGHT ANY HOLES MADE FOR GALVANISING.
5. THE CONTRACTOR SHALL PROVIDE ALL CLEATS AND HOLES NECESSARY FOR FIXING STEEL TO STEEL AND TIMBER TO STEEL, WHETHER OR NOT DETAILED ON THE STEELWORK DRAWINGS.
6. UNLESS SPECIFIED OTHERWISE, WELDS SHALL BE GENERAL PURPOSE (GP) CONTINUOUS FILLET WELDS. DEFAULT SIZE IS LESSOR OF 6mm AND THICKNESS OF THE THINNEST PART BEING JOINED. STRUCTURAL PURPOSE (SP) WELDS SHALL BE SELECTED FOR STRUCTURES SUBJECT TO FATIGUE.
7. ANY CUTS, HOLES AND WELDS TO EXISTING STEELWORK SHALL BE TREATED WITH 'COLD-GAL' ZINC RICH PAINT.
8. BOLTS NOT DESIGNATED SHALL BE GRADE 8.8 TO AS/NZS 1252 AND 'SNUG' TIGHTENED.
9. ALL BOLTS AND U-BOLTS SHALL BE SNUG TIGHTENED AND SECURED WITH FLAT WASHER AND SPRING WASHER (UNLESS SPECIFIED OTHERWISE).
10. PROVIDE APPROVED NEOPRENE (OR EQUIVALENT) WASHERS AND COLLARS AT ALL DISSIMILAR METAL INTERFACES.
11. ALL CHEMICAL AND MECHANICAL MASONRY ANCHORS SHALL BE STAINLESS STEEL (UNLESS SPECIFIED OTHERWISE) AND INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S SPECIFICATION.
12. TWO-HOLE EARTH LUGS SHALL BE PROVIDED ON ALL MOUNTS, EXCEPT ON STEEL MONOPOLES, LATTICE TOWERS AND GUYED MASTS.

1. ALL CONCRETE WORK SHALL COMPLY WITH THE CURRENT ISSUE OF AS 3600 AND OTHER RELEVANT STANDARDS REFERENCED THEREIN.
2. UNLESS SPECIFIED OTHERWISE, CONCRETE MIX SHALL CONTAIN PORTLAND CEMENT TYPE GP OR GB, 20mm GRADED COARSE AGGREGATE AND ACHIEVE 80mm SLUMP AT POINT OF DELIVERY.
3. UNLESS SPECIFIED OTHERWISE, CONCRETE COMPRESSIVE STRENGTH AT 28 DAYS (f'_c) SHALL BE 32 MPa.

1. ALL MASONRY WORK SHALL CONFORM WITH THE CURRENT ISSUE OF AS 3700 AND OTHER STANDARDS REFERENCED THEREIN.
2. BUILD IN AS NECESSARY, LINTELS, FRAMES, BOLTS, LUGS, WALL TIES AND METALWORK.
3. CAREFULLY POSITION OPENINGS FOR OTHER TRADES TO ELIMINATE CUTTING.
4. BED JOINTS ARE TO BE 10mm THICK.
5. BEFORE LAYING MASONRY UNITS, ENSURE THAT THE BASE IS CLEAN AND FREE OF LAITANCE.
6. INSTALL WIRING FOR POWER AND OTHER CONDUITS WITHIN BLOCK CORES (WHERE APPLICABLE). DO NOT CUT CHASES IN HOLLOW BLOCKWORK.
7. ALL WALL INTERSECTIONS SHALL BE OF BONDED CONSTRUCTION OR TIED TO EXISTING WITH MEDIUM DUTY (MIN) TIES AT 400mm CENTRES VERTICALLY.

1. THE ELECTRICAL INSTALLATION SHALL COMPLY WITH THE RELEVANT CODES, STANDARDS, RULES, AND REGULATIONS OF STATUTORY AUTHORITIES. IN PARTICULAR: STATE SERVICE AND INSTALLATION RULES, AS/NZS 3000 WIRING RULES, AS3015, AS/NZS 3017 AND THE SUPPLY AUTHORITY REGULATIONS.
2. THE LOCATIONS AND MOUNTING HEIGHTS OF THE NEW INSTALLATION SHOWN ON THE SITE DRAWINGS IS INDICATIVE ONLY. THE FINAL LOCATIONS AND HEIGHTS SHALL BE DETERMINED ON SITE TO SUIT THE CLIENT AND THE INSTALLATION. THE CONTRACTOR SHOULD VISIT THE SITE DURING THE TENDER PERIOD TO BECOME FAMILIAR WITH THE SCOPE OF WORK.
3. ALL REQUIREMENTS FOR INSTALLATIONS REGARDING SUPPLY ARRANGEMENTS, PROVISION OF SERVICE CABLES AND CONSUMER MAINS AND METERING SHALL BE IN ACCORDANCE WITH AS/NZS 3000 AND RELEVANT STATE SERVICE AND INSTALLATION RULES.
4. A SEPARATE MEN EARTH ELECTRODE SHALL BE PROVIDED BY THE CONTRACTOR BELOW THE METER BOX AS REQUIRED.
5. ALL WIRING SYSTEMS SHALL BE INSTALLED AND ENCLOSED BY APPROVED METHODS WHICH WILL READILY PERMIT CABLES TO BE DRAWN IN OR REPLACED AFTER COMPLETION OF CONSTRUCTION.
6. CABLE MARKERS SHALL BE PROVIDED FOR UNDERGROUND WIRING FROM WITHIN THE PROPERTY BOUNDARY AND TO THE OPTUS INSTALLATION, AT THE COMMENCEMENT AND FINISH OF EACH ROUTE AND AT EACH CHANGE OF DIRECTION.
7. ALL EXISTING SURFACES, KERBS, GUTTERS, INVERTS, VEHICLE CROSSINGS AND PAVEMENTS DISTURBED AFTER INSTALLATION OF UNDERGROUND WIRING SHALL BE REINSTATED AND MADE GOOD BY THE CONTRACTOR.
8. LOCATE AND IDENTIFY ALL UNDERGROUND SERVICES BEFORE COMMENCING WORK.

1. EQUIPMENT AND ANTENNA MOUNTS SHALL BE EARTHED IN ACCORDANCE WITH THE EARTHING SPECIFICATION (OSD-030).
2. ALL EXTERIOR EARTH TAPE ELECTRODES SHALL BE COPPER BONDED HARDENED STEEL UNLESS OTHERWISE NOMINATED ON THE SITE DRAWINGS.
3. THE EARTHING SYSTEM SHALL BE CHECKED FOR CONTINUITY AND IMPEDANCE SHALL BE MEASURED AND A WRITTEN TEST REPORT PROVIDED.
4. EARTHING ELECTRODES SHALL BE INSTALLED AT A DEPTH OF NOT LESS THAN 3 METRES UNLESS NOTED OTHERWISE.

1. EXISTING SERVICES SHOWN ON SITE DRAWINGS IS REPRESENTATIVE OF AVAILABLE INFORMATION (OBTAINED FROM LOCAL AUTHORITIES) AND THE SITE SURVEY.
2. SERVICES INFORMATION SHALL BE READ IN CONJUNCTION WITH THE RELEVANT LOCAL AUTHORITIES DRAWINGS TO CONFIRM ACCURACY AND COMPLETENESS.
3. ADDITIONAL UNDOCUMENTED SERVICES MAY BE PRESENT ON SITE. FOR INFORMATION OF UTILITY UNDERGROUND SERVICES CALL 1100 'DIAL BEFORE YOU DIG'.
4. THE CONTRACTOR SHALL IDENTIFY AND CONFIRM THE LOCATION OF ALL RELEVANT UNDERGROUND SERVICES PRIOR TO COMMENCEMENT OF THE WORKS USING MANUAL POTHOLING OR OTHER APPROVED MEANS.



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OPTUS

MOBILE NETWORK
AUSTRALIA

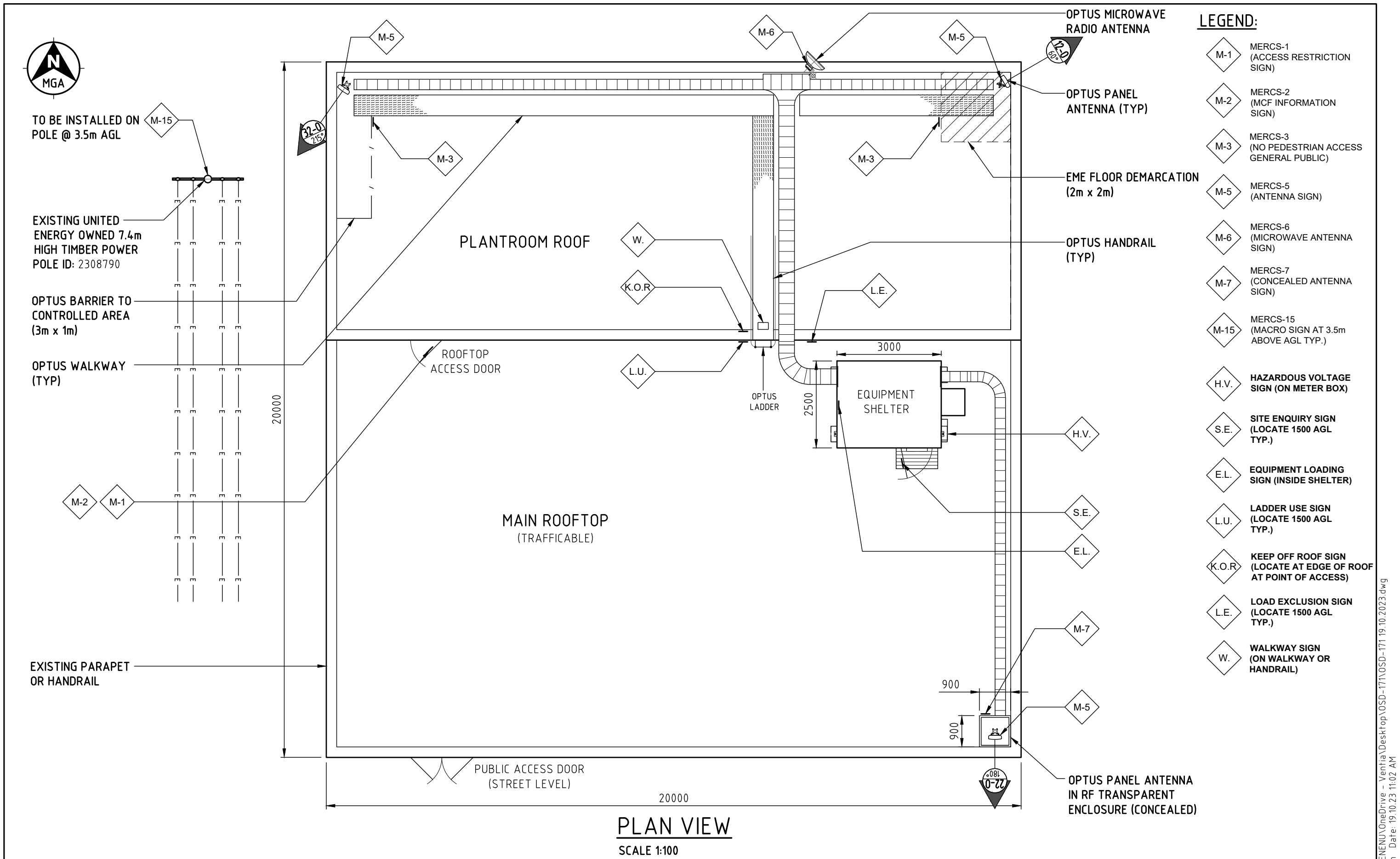
OPTUS STANDARD
DRAWING

STANDARD CONSTRUCTION NOTES

Drawing Status:
FOR CONSTRUCTION

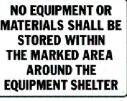
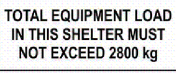
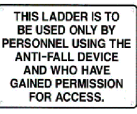
Drawing No.
OSD-100

Revision
C



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SIGNAGE LEGEND

SIGN NAME	SYMBOL	SIGN IMAGE	SIGN NAME	SYMBOL	SIGN IMAGE	SIGN NAME	SYMBOL	SIGN IMAGE	SIGN NAME	SYMBOL	SIGN IMAGE
MERCS-1 (ACCESS RESTRICTION SIGN)	M-1		MERCS-9 (IN-BUILDING ANTENNA SIGN)	M-9		MERCS-19 (SMALL CELL ISOLATION SWITCH SIGN AFFIXED NO HIGHER THAN 4m AGL TYP.)	M-19		OPTUS ISOLATION SWITCH (MICROCELL RF SHUTDOWN SIGN)	O.I.S.	
MERCS-2 (MCF INFORMATION SIGN)	M-2		MERCS-10 (REPEATER ANTENNA SIGN)	M-10		MERCS-20 (GENERIC ADJACENT RF HAZARD SIGN)	M-20				
MERCS-3 (NO PEDESTRIAN ACCESS GENERAL PUBLIC SIGN)	M-3		MERCS-14 (ISOLATION SWITCH SIGN)	M-14		HAZARDOUS VOLTAGE SIGN ON METER BOX	H.V.				
MERCS-4 (NO PEDESTRIAN ACCESS OCCUPATIONAL SIGN)	M-4		MERCS-15 (MACRO SIGN AT 3.5m ABOVE AGL TYP.)	M-15		SITE ENQUIRY SIGN (LOCATE 1500 AGL TYP.)	S.E.				
MERCS-5 (ANTENNA SIGN)	M-5		MERCS-16 (MACRO SIGN)	M-16		KEEP OUT SIGN (LOCATE 1500 AGL TYP.)	K.O.				
MERCS-6 (MICROWAVE ANTENNA SIGN)	M-6					KEEP OFF ROOF SIGN (LOCATE AT EDGE OF ROOF AT POINT OF ACCESS)	K.O.R.				
MERCS-7 (CONCEALED ANTENNA SIGN)	M-7		MERCS-17 (STREET LIGHT SIGN)	M-17		LOAD EXCLUSION SIGN (LOCATE 1500 AGL TYP.)	L.E.				
MERCS-8A (MICROCELL SIGN 2m)	M-8A					EQUIPMENT LOADING SIGN (LOCATE 1500 AGL TYP.)	E.L.				
MERCS-8B (MICROCELL SIGN 1m)	M-8B		MERCS-18 (SMALL CELL SIGN)	M-18		LADDER USE SIGN (LOCATE 1500 AGL TYP. TYP.)	L.U.				
MERCS-8C (MICROCELL SIGN 0.5m)	M-8C					WALKWAY SIGN (ON WALKWAY OR HANDRAIL)	W.				

- NOTES:
- REFER TO CURRENT RFSP, MANUAL 2 "PREFERRED SITE SIGNAGE" FOR EME SIGNAGE REQUIREMENTS AND INSTALLATION REQUIREMENTS.
 - REFER TO SECTION 5 OF THE OPTUS MOBILE NETWORK DRAWING SPECIFICATION (OSD-070) FOR DRAWING REQUIREMENTS.
 - REFER TO SECTION 6 OF THE OPTUS DESIGN & CONSTRUCTION SPECIFICATION (OSD-010) FOR SIGNAGE DETAILS INCLUDING SIZE, MATERIALS AND FIXING.
 - THIS IS A TEMPLATE DRAWING ONLY. ACTUAL SITE DRAWING MUST BE PRODUCED TO REFLECT SITE SPECIFIC CONDITION.

Rev	Date	Revision Details	Consultant	CAD	Designer	Verifier	Approver
B	10.10.23	SMALL CELL DRAWING REMOVED	VENTIA	UH	RY	-	-
A	05.04.22	REVISED AND CONSOLIDATED OSD-170, OSD-180 AND OSD-191 INTO THIS DRAWING ADDED NEW TYPICAL SMALL CELL DRAWING DETAILS OF THE CHANGES AND REQUIREMENTS ARE IN OSD-070 SECTION 5.14	KORDIA	UH	RY	-	-



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Client:
Project:
MOBILE NETWORK AUSTRALIA
SITE No:- OPTUS
STANDARD DRAWING

Drawing Title:
SIGNAGE LEGEND AND NOTES

Drawing Status:
FOR CONSTRUCTION

Drawing No.
OSD-171-3

Revision
B